

Curbing Leakage in Public Programs: Evidence from India's Direct Benefit Transfer Policy[†]

By PRABHAT BARNWAL*

Targeted price subsidies create a gap between subsidized and unsubsidized prices. The resulting dual pricing can lead to arbitrage opportunities where intermediaries divert subsidized goods to unintended beneficiaries via the black market. I study India's Direct Benefit Transfer policy for cooking fuel subsidies, which altered the existing subsidy program by transferring subsidies directly to beneficiaries' bank accounts. The policy decreased subsidized fuel purchases, indicating a reduction in diversion to the black market. Changes in unsubsidized fuel sales and black market prices provide supporting evidence that leakage was reduced. These results suggest that addressing the underlying perverse incentives in welfare delivery can improve efficiency by curbing leakages. (JEL D73, I38, O17, Q41, Q48)

The design of welfare programs in many low- and middle-income countries (LMICs) includes a combination of targeting and price subsidies, whereby prices of essential commodities such as food and fuel are subsidized for targeted beneficiaries (World Bank 2015). A common by-product of such a design is the creation of a dual-price system whereby program beneficiaries pay lower prices for the *same* commodity that nonbeneficiaries buy at a higher price on the open market. This creates incentives for leakage and corruption by making it lucrative for intermediaries to divert subsidized commodities to the open market. Such leakage hurts government finances and may distort (inflate) open market purchases of the commodity.

This problem is exemplified by India's approach to subsidizing cooking fuel for households till 2013. Households were allowed to buy Liquefied Petroleum Gas

*Michigan State University (email: prabhat@msu.edu). Luigi Pistaferri was the coeditor for this article. I thank the three anonymous referees for their constructive comments throughout the review process. I am particularly grateful to Neeraj Mittal and Ashutosh Jindal at the Ministry of Petroleum and Natural Gas, India, for sharing data and helpful discussions. I thank Wojciech Kopczuk, Cristian Pop-Eleches, and Eric Verhoogen at Columbia University for their guidance. I cannot thank Chris Ahlin enough for reading the draft multiple times and providing many useful suggestions. I thank Douglas Almond, Soren Anderson, Emily Breza, Ritam Chaurey, Manasi Deshpande, Stacy Dickert-Conlin, Jonathan Dingel, Ram Fishman, Raymond Fisman, Francois Gerard, Siddharth George, Jessica Goldberg, Michael Greenstone, Jonas Hjort, Supreet Kaur, Dilip Mookherjee, Karthik Muralidharan, Suresh Naidu, Arvind Panagariya, Rathin Roy, Nicholas Ryan, Bernard Salanié, Pronab Sen, Arvind Subramanian, Wolfram Schlenker, Enrique Seira, Sandip Sukhtankar, Anna Tompsett, Jan von der Goltz, Ben Zou, and various seminar participants for helpful comments. Shyam Peri, Abhinava Prakash, Abhishek Beriya, and Xin Liu provided outstanding support with the data and fieldwork. I gratefully acknowledge financial support from the IGC (1-VCS-VINC-VXXXX-35113, 1-VRE-VCCN-VXXXX-89111), CIBER, and Dr. Delia Koo Endowment at MSU. This research was reviewed by Columbia University IRB (AAAM8905). All errors are mine. I declare that I have no relevant or material financial interests that relate to the research described in this paper.

[†]Go to <https://doi.org/10.1257/aer.20161864> to visit the article page for additional materials and author disclosure statement(s).

(LPG, commonly known as ‘cooking gas’ or ‘gas’) at a *subsidized price*, up to a generous annual cap; and at a tax-free *unsubsidized price* beyond the cap. Commercial users faced a tax-inclusive *commercial price* that was significantly higher than either household price. This design was susceptible to diversion of the lower-price gas meant for households to commercial users. Intermediary agents, including gas distributors in the public distribution system, are thought to have used ghost and duplicate accounts to obtain household gas cheaply, reselling it to commercial users via black markets. They also allegedly made extra purchases from legitimate household accounts. The fact that in 2011–2012 the number of household gas accounts was about 50 percent higher in the Petroleum Ministry’s administrative records than in the population census provides suggestive evidence for the existence of fake accounts. It appears that government enforcement policy, based on audits and fines, was largely ineffective.

In 2013, the government of India introduced Direct Benefit Transfer for LPG (DBTL), a reform whereby the over-the-counter price subsidy for gas was replaced with a direct transfer of the subsidy into beneficiary bank accounts. The policy reform changed only the mode, but not the level, of subsidy provision. Under DBTL, a household purchased gas at the unsubsidized price from a distributor, the transaction was recorded electronically, and the subsidy amount was directly deposited to the household’s bank account within days of purchase. To enroll in DBTL, households had to submit their *Aadhaar* (a 12-digit biometric ID) and bank account details. *Aadhaar* and bank account details together made it feasible for the government to make bank transfers of the subsidy to legitimate households.

DBTL was implemented across districts in a staggered manner, but was unexpectedly repealed in March 2014.¹ I study the impact of this reform with a difference-in-differences methodology using both the rollout and the repeal of the DBTL policy. In essence, I first compare districts where DBTL was mandated earlier to districts where it was not yet mandated, during the *policy rollout* in 2013–2014. Next, I compare districts where the policy was mandated to those where it was not, before and after its *repeal* in March 2014. The government selected districts for earlier DBTL implementation based on district-level *Aadhaar* enrollment in early 2013, which is plausibly exogenous to cooking gas subsidy leakage. Indeed, data from the pre-reform period shows similar trends in gas sales across treatment and control districts, supporting the parallel pre-trends assumption.

I use administrative data on distributor-month-level sales of gas to household and commercial accounts covering January 2013–August 2015. I also use transaction-level information on household gas (i.e., gas purchased from the distributor by household accounts) during the 2013–2014 fiscal year: 23 million transactions carried out by around 4 million household gas accounts. Data on commercial gas (i.e., gas purchased from the distributor by commercial accounts) includes distributor-month-level information on a type of commercial gas refill (19kg units) mostly used by small businesses. Additionally, I conducted surveys to collect black market price information from gas deliverymen and commercial users. I solicited

¹ The incumbent government repealed the policy in the run-up to national elections, citing implementation-related issues as the reason, though political lobbying by vested interests may also have played a role. The newly elected government restored the reform in late 2014.

black market prices from deliverymen in an audit survey with enumerators acting as customers, and complement this data with prices collected from small businesses. This fieldwork was carried out in 89 districts in 11 states, and covered the periods immediately before and after the policy's termination.

My results show a reduction of about 17–19 percent in total gas sales to household accounts when DBTL was enforced. This divergence between DBTL and non-DBTL districts vanishes after the repeal. Assuming that *genuine* households did not dramatically decrease gas purchases in response to the policy change, these results suggest that it led to a substantial decrease in purchases by *fake* household accounts, that is, to a significant reduction in leakage.

Further evidence is found in the commercial gas market. Though the commercial gas price was not directly affected by the DBTL policy, I estimate that the amount of gas sold by distributors to commercial accounts (in 19kg units) increased by about 13–14 percent in response to the policy change. As with household gas, commercial gas sales in DBTL districts reverted to the same level as in non-DBTL districts when the policy was repealed. That DBTL induced commercial users to substitute toward buying gas through the formal channel at a higher price strongly suggests a negative supply shock in the black market for diverted household gas.

Black market data provide evidence in support of a diversion-based supply shock: I estimate that the termination of DBTL brought down black market prices by about 18–20 percent.² Taken together, these results suggest that DBTL caused a negative supply shock in the black market by curtailing diversion, while its repeal caused a positive supply shock by reenabling diversion.

A substantial decline in diversion would be a welcome outcome; a more worrying possibility is that some of the decline in subsidized household purchases under DBTL represented the exclusion of legitimate households from accessing the subsidy, due perhaps to the lack of a bank account or an Aadhaar. The possibility for exclusion is a known byproduct of increased complexity in welfare delivery programs (Kleven and Kopczuk 2011; Deshpande and Li 2019; Finkelstein and Notowidigdo 2019). While this study cannot rule out some exclusion, it does suggest that leakage reduction and not exclusion is the main driver of results. First, the behavior of commercial gas purchases and black market prices discussed above is not easily explained by exclusion and argues for significant leakage reduction. Second, descriptive statistics suggest that Aadhaar coverage and banking access are not binding constraints for most household gas consumers, likely due to the fact that cooking gas is primarily a fuel used by urban middle- to higher-income households during the study period. Third, using data from an independently conducted household survey, I do not find much impact of DBTL on the average share of households' cooking fuel expenditure. Finally, I estimate heterogeneous impacts of DBTL on subsidized household gas purchases by initial purchase levels, and find that impacts are greater among higher-initial-purchase household accounts. Taken together, the evidence suggests that substantial leakage reduction occurred, and that comparatively, any exclusion of genuine households was a less important factor.

²Black market price data were collected before and after the repeal of mandatory DBTL. This means I can only estimate the impact of DBTL termination on black market prices, and not of DBTL implementation.

A more detailed view comes from comparing magnitudes of changes in gas purchases at each of the three prices. For every DBTL-induced 1 kg decline in household purchases at the subsidized price, household purchases at the unsubsidized price rose 0.68 kg while commercial purchases (of 19 kg units) rose 0.05 kg. With the caveat that an unknown amount of commercial purchases (when sold in units other than 19 kg) is missing in this analysis, this suggests that most of the leakage reduction came from a black market shift in sourcing from subsidized gas to unsubsidized but untaxed gas. This is unsurprising since DBTL affected receipt of the subsidy, but not access by household accounts to the unsubsidized price; as this price was still substantially below the commercial price, arbitrage opportunities remained. Overall, the estimates are consistent with a leakage-reduction story: after DBTL is implemented, agents could only divert more expensive (i.e., unsubsidized) household gas; large-scale commercial purchases from the black market continued, alongside small-scale substitution toward legally purchased, expensive commercial gas; and overall gas consumption by commercial users declined as the gas had become more costly for them. And while the subsidy leakage appears to have been significantly reduced, there remained a substantial tax-related leakage that DBTL was not designed to address.

The first and most important contribution of this paper is to the literature on the design and delivery of welfare programs, especially in LMICs. Specifically, I provide empirical evidence documenting how the dual price system created by targeted subsidies can lead to leakage and diversion of public funds. I also show how replacing such subsidies with direct income transfers can reduce corruption and leakage. This possibility has been widely conjectured in prior work (Alderman 2002; Kojima 2013; Kotwal, Murugkar, and Ramaswami 2011; Beegle, Coudouel, and Monsalve 2018; Khera 2011a), and this paper provides well-identified evidence to validate this idea.

Second, the paper demonstrates the value of modern technologies such as biometric identification and bank accounts in expanding state capacity in LMICs. Prior work such as Muralidharan, Niehaus, and Sukhtankar (2016) and Banerjee et al. (2020) has shown how such payments infrastructure can improve the implementation of existing programs. My paper shows how modern payments infrastructure can expand the feasible set of policies that governments in LMICs can implement. Specifically, the reform studied in this paper would not have been possible to implement without the government of India's investments in Aadhaar and bank accounts for citizens. In a similar vein, Banerjee et al. (2023) show how using electronic food vouchers to transfer benefits allowed the government of Indonesia to improve the targeting of benefits by replacing in-kind transfers with income transfers.

Finally, the paper contributes to the literature on detecting corruption. Since data on corruption and black market activities are often difficult to obtain, methodological advances in the measurement of corruption are of broad interest beyond the specific context studied. Examples of studies that use novel approaches to measure corruption and black market activities include Fisman and Wei (2004); Pissarides and Weber (1989); Fisman and Wei (2009); and Olken (2007). Studying changes in black market prices as a proxy for changes in corruption (as done in this paper) is a method that could also be used in other settings to detect the diversion of subsidized commodities.

From a policy perspective, the success of the reform studied in this paper points to the broader applicability of a similar approach to reforming other targeted in-kind subsidies that are prone to diversion. A similar combination of targeting and price subsidization is commonly seen in the design of massive public distribution systems (PDS) in LMICs: for example, for food and fertilizer, creating incentives for leakage and corruption. This paper shows that it is possible to reduce leakage in the PDS by changing the mode of subsidy delivery from over-the-counter to direct bank transfer.³

The rest of the paper proceeds as follows. Section I discusses the institutional context and the policy reform. Data and the empirical strategy are discussed in Sections II and III, respectively. Section IV reports the results, and Section V concludes.

I. Institutional Background and Policy Details

India's cooking gas subsidy program offers two unique strengths for this study. First, gas is a homogenous good. This allows abstraction from any concerns about quality differences between in-kind transfers and the open market (Besley and Coate 1991). The assumption that goods provided to beneficiaries by the state at a subsidized price are of the same quality as goods purchased by nonbeneficiaries in the open market is unlikely to be true in most in-kind transfer programs (Balakrishnan and Ramaswami 1997; Jacoby 1997). Second, the base price of gas (i.e., the unsubsidized household gas price) is regulated by the Indian government in tandem with plausibly exogenous global prices and adjusted each month, while the subsidized price remains constant over the period of interest. Thus, local demand and supply factors do not affect official subsidized or unsubsidized gas prices for users.

A. Public Distribution System for Cooking Gas

Three oil marketing companies (OMCs), controlled by the Indian Ministry of Petroleum and Natural Gas, supply gas to households and commercial consumers such as restaurants and businesses across the country. In 2012–2013, the household gas share was about 88 percent of all gas sold by OMCs, with the remaining 12 percent consumed by the commercial, industrial, and transportation sectors (Ministry of Petroleum and Natural Gas 2014b). All three OMCs are simultaneously present across the country through a local franchise system known as the gas distributorship.

Gas refill transactions allow households and commercial users to exchange empty cylinders for full ones. Throughout this paper, I refer to the gas refills sold by distributors to household accounts as “household gas” and to commercial accounts as “commercial gas.” The median household consumes about seven 14.2 kg refills per year, meaning that many households can go for several months without refilling even one gas cylinder. OMC distributors sell gas to household accounts in 14.2 kg

³Olken and Pande (2012) review the empirical literature on corruption in government provision of goods and services in developing countries. Even moderate rates of leakage may outweigh the potential welfare gains from in-kind transfer programs (Khera 2011a; Gadenne 2018). Recent papers highlight the role of information (Banerjee et al. 2018) and informal monitoring (Nagavarapu and Sekhri 2016) in reducing leakages in in-kind transfer programs.

cylinders and to commercial accounts in 19 kg cylinders. Commercial users can also buy gas in larger size cylinders or in bulk directly from commercial-only gas distributors. Also, a different set of retailers sell gas for vehicle fuel usage. This means that 19 kg gas refills represent only a subset of commercial gas sales. Except for a few urban areas with access to piped gas for households, nearly all household gas is sold in 14.2 kg cylinders.

Historically, political connections and bribes have played an important role in the allotment of gas distributorships, gas stations, and food PDS ration shops in India (Chakravarty and Joshi 1997; Banerjee and Pande 2007). Gas distributorships are initially licensed for 10 years with the possibility of renewal; in practice, almost all licensees are regularly renewed.

B. Pricing and Subsidy

Gas is subsidized for household cooking use only, while it is taxed for commercial users and when households want to use it for other purposes (e.g., as vehicle fuel). The policy rationale is to encourage gas usage, discourage highly polluting solid fuel usage for cooking and shield households from international petroleum price fluctuations. While gas subsidies are universally available to all households, mostly middle- to higher-income households end up using gas, and thus, disproportionately availing the gas subsidy. This is because even the subsidized gas and the associated setup costs are often beyond the means of the low-income households. Households with low income, especially those living in rural areas, either stack gas with other fuels or not use it at all (Gould and Urpelainen 2018). National Sample Survey data from 2011–2012 indicates that 60 percent of the total subsidy value going to households in the top income quintile, while the bottom two quintiles received almost none of the subsidy (Figure 1).⁴ The household take-up of gas as the primary cooking fuel was 30 percent in 2011 (census), which improved to 42 percent in 2015 (IIPS and ICF 2017), suggesting an increase in adoption among lower-income households.

Cooking gas is sold by the OMCs at three different prices: the subsidized household gas price ($p - s$), the unsubsidized household gas price (p), and the commercial gas price ($p + t$). The unsubsidized household gas price p is similar to a base *regulated price* that is revised by the government on a monthly basis to track (roughly) the international market price. However, the government deliberately keeps the subsidized price ($p - s$) stable using a floating subsidy amount s , making the subsidized gas price households face effectively constant over time. The commercial gas price equals the unsubsidized household gas price p plus taxes t . The additional tax on commercial gas t consists mostly of excise and value-added taxes. Figure 2, panel A illustrates the gas pricing structure using price data from January 2014. The regulated price (unsubsidized household price) was about 1.5 US\$ per kilogram. Households received a 66 percent subsidy while commercial users faced 22 percent in additional taxes on the regulated price.

⁴By contrast, the take-up of rice and wheat subsidies in the lowest consumption class (i.e., 0–5 percent fractile class of monthly per capita consumer expenditure) in urban India was 32 percent and 23 percent, respectively (NSSO 2015b).

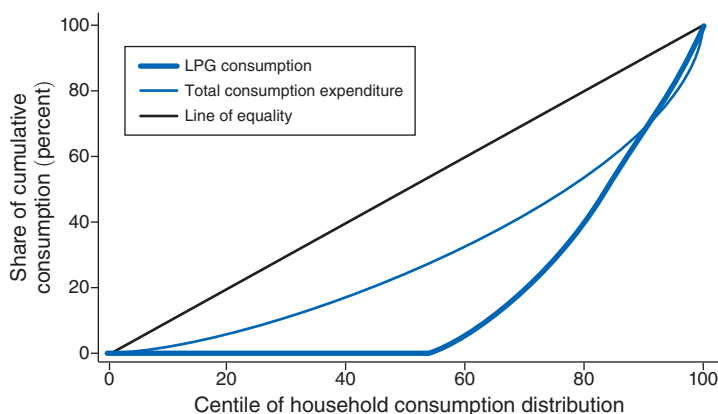
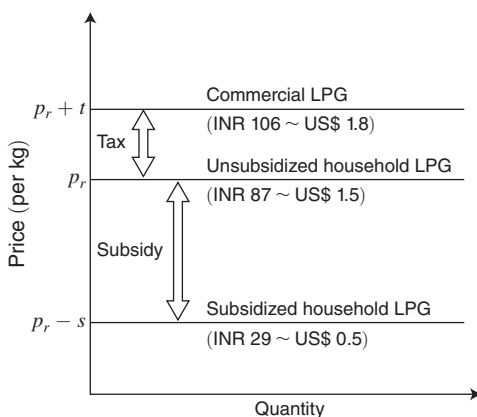


FIGURE 1. LPG SUBSIDY TARGETING

Notes: This graph shows the inequality in household gas purchase using NSS 68th round (2011–2012) household survey data. The 45-degree line shows the equal distribution, while the light blue curve shows the distribution of total expenses, as a proxy for income. The bold-blue curves shows the distribution of household gas purchase.

Panel A. LPG pricing



Panel B. Reported versus actual beneficiaries

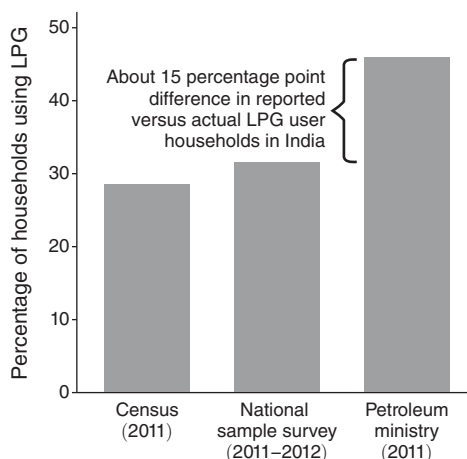


FIGURE 2. LPG SUBSIDY PROGRAM

Notes: Panel A shows prices for subsidized household gas, unsubsidized household gas, and commercial gas refills. p denotes the unsubsidized household gas price. Households face subsidized price ($p - s$), whereas commercial firms purchase fuel at a tax inclusive price ($p + t$). The subsidy s is floating so that subsidized household gas price $p - s$ remains stable for households. Household accounts that exceed the annual cap on refills or household accounts that did not enroll in DBTL can continue buying household gas refills at unsubsidized price p . January 2014 prices for Delhi approximated in USD are shown using the 2014 exchange rate of 1 US\$ = ₹60. 14.2kg unsubsidized household gas refill was available for ₹1,241, while a subsidized gas refill of the same size would cost ₹414 with ₹827 subsidy. Nineteen kg commercial refill was priced ₹2,014 after taxes. Local prices may further vary due to variations in transportation costs and state-specific taxes.

Panel B: This graph shows the percentage of households reporting that they use gas for cooking from the 2011 Census and 2011–2012 National Sample Surveys, as well as the percentage of households using cooking gas in the administrative data (Ministry of Petroleum and Natural Gas 2011; Registrar General of India 2011; NSSO 2015a). All three sources consider a similar base number of the total number of households (~250 million) in 2011.

To purchase gas refills, households need to open an account at the local distributorship for a net cost of about 6 US\$ (about ₹360 in 2014). Each household account can buy subsidized refills up to an annual cap. In 2013, household accounts were entitled to up to nine 14.2 kg refills. Later the government allowed households to purchase one additional subsidized refill each in February and March 2014. From 2014–2015 onwards, households can purchase up to twelve subsidized 14.2 kg refills in a year. Households can always purchase gas beyond the cap at the unsubsidized household gas price. The annual cap of nine 14.2 kg gas refills at the time of DBTL implementation was exceeded by less than 5 percent of household accounts, as per the administrative data.

C. Enforcement of Dual Pricing and Diversion Incentives

The government enforces market segmentation between household and commercial gas accounts with audits and penalties. This enforcement hinges on two main factors. First, visual differentiation in color, shape, and size of household and commercial gas cylinders helps reduce enforcement costs, similar to the red dye in untaxed diesel in the United States (Marion and Muehlegger 2008). However, this is not a foolproof strategy when monitoring is poor. Second, it is illegal for unauthorized agents to sell, transport, or refill gas. In practice, the audit and penalty policies have not been very effective in curbing diversion since enforcement is costly, requires significant coordination among agencies, and is susceptible to local-level corruption and side payments. The Indian government faced difficulties countering these financial incentives for diversion using fines and raids. Figure 2, panel B confirms this: the number of household LPG accounts was significantly higher in the administrative record than in the household census and national sample surveys in 2011–2012, suggesting large-scale diversion using ghost and duplicate accounts in practice.⁵

The role of gas distributors and their deliverymen becomes pivotal in the black market. Since they deliver household as well as commercial gas refills, they have the ability to divert household gas through the creation of “ghost” accounts or by illicitly taking refills from legitimate household accounts. Independent agents and cartels can also purchase household gas through ghost accounts and resell it in the black market, although any large-scale operation will not be feasible without tacit collusion with distributors and deliverymen. Audit results suggest that the number of fraudulently created gas subsidy beneficiaries may run into the millions (Sastry 2012). On the intensive margin, distributors may also over-report refills purchases from legitimate household accounts to divert subsidized gas to the black market. The Essential Commodities Act imposes severe legal penalties on independent agents and cartels active in the black market for gas. However, it is practically impossible to punish the distributor or deliverymen complicit in black market transactions since they are authorized to carry, store, and sell both household and commercial types of

⁵Fuel subsidy programs in many LMICs face similar challenges. Fuel subsidies are distorting and costly, especially when they are cornered by nonintended beneficiaries (McLure Jr 2014; Arze del Granado, Coady, and Gillingham 2012). Evasion of fuel subsidies and taxes in LMICs remains relatively unexplored in the empirical literature (Kojima 2013).

gas cylinders, and there is a high degree of tolerance for irregularities in distributor accounting records.

Other than distributors and private cartels, legitimate households may also resell household gas from their own quota to make a profit. While the diversion by legitimate households cannot be ruled out, it would be more costly for a household to resell gas than for a distributor or deliverymen. The cap on household subsidized fuel purchases leaves a limited margin for resale beyond the household's own consumption. Further, the potential risk of getting detected, as well as the search cost for black market transactions, may be higher for households.

D. *Direct Benefit Transfer of LPG Subsidy (DBTL)*

Under DBTL, subsidy benefits are transferred directly into the bank accounts of verified household beneficiaries, conditional on gas refill purchase. Unlike many other cash transfer programs, DBTL is not a lump sum transfer system. Under this policy, households continue to face effectively the same lower price net of the subsidy transferred to their bank account. However, the large *over-the-counter price gap* between household gas and commercial gas shrinks under DBTL, compared to the system before DBTL.

In order to enroll in DBTL, households need to provide bank account information and an Aadhaar number. India provides unique Aadhaar numbers to residents after collecting their name, address, date of birth, gender, photograph, proof of identity, proof of address, and biometric data (iris scans and fingerprints) at designated centers. As soon as a household enrolls in DBTL, a cash advance worth the subsidy for one gas refill is transferred to the bank account. This advance provision exists to address issues related to liquidity constraints that may lead to exclusion of the poor. In the data used in this paper, the average duration between gas purchase and subsidy transfer was about three days.

As shown in Figure 2, panel A, there were three price levels before DBTL: the unsubsidized household gas price p , subsidized household gas price $p - s$, and commercial gas price $p + t$, where s is the subsidy provided to household accounts and t is the additional tax imposed on gas sold to commercial accounts. DBTL did not change the effective gas price for legitimate household accounts enrolled in DBTL. For subsidized gas refills under DBTL, household accounts paid p to the gas distributor for gas refills, and the subsidy amount s was transferred directly to the purchasing household's bank account. Irrespective of enrollment in DBTL, all household accounts could purchase unsubsidized household gas at price p . Thus, by eliminating the over-the-counter provision of household subsidy, DBTL bridged the price gap between commercial and household gas to a significant extent.⁶ However, this policy reform did not entirely eliminate the price gap, since unsubsidized household gas still remained cheaper than commercial gas due to the additional taxes on gas sold to commercial accounts.

Households could choose to enroll in DBTL when they needed gas refills. In Phase 1 districts (described below), about 60 percent household accounts had

⁶This draws parallels with the policy reforms that change the point of commodity tax collection to reduce evasion (Slemrod 2008; Kopczuk et al. 2016).

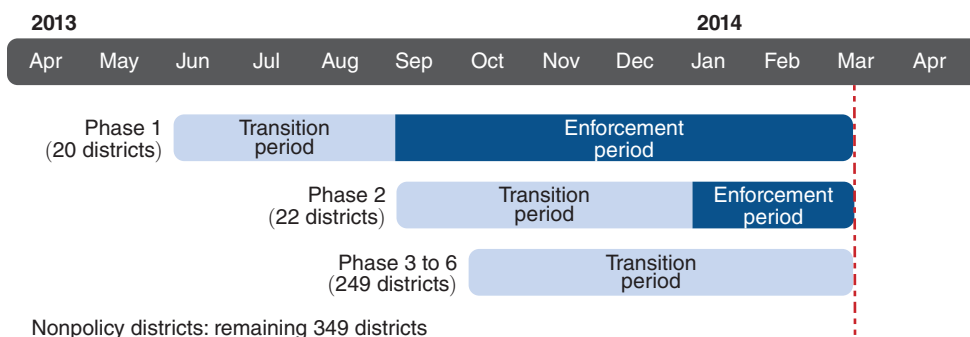


FIGURE 3. TIMELINE OF DBTL POLICY ROLLOUT AND TERMINATION

DBTL Rollout: DBTL is first introduced for voluntary enrollment and after the transition period is over, it is subject to mandatory enforcement. In the voluntary DBTL period, households enrolled in DBTL receive direct transfers while all other household LPG users (including ghost accounts) continue to receive household gas refills as usual. Once DBTL becomes mandatory, all household gas users purchase gas only at the unsubsidized price, and only those households enrolled in DBTL receive direct subsidy transfers after their refill purchase. By March 2014, the DBTL policy was available on a voluntary basis in 291 districts, but it was only fully enforced in 18 districts from Phase 1 and 22 districts from Phase 2. Two districts from Phase 1, Mysore and Mandi, observed a one-month and two-month delay, respectively, in the enforcement of DBTL due to by-elections. 14 additional districts from Kerala state were included in Phases 1 and 2 for voluntary DBTL, but DBTL was not enforced there until its termination in March 2014.

Policy Termination: Policy termination effectively restored the status quo subsidy delivery system.

Data Coverage: Administrative data on distributor-month level sales covering household and commercial fuel across the country is available from January 2013 to August 2015. The household-month level panel is available for about 10 percent of household gas users from HPCL for one full financial year 2012–2013 (April 2013–March 2014). Black market survey data collection was done before and after the DBTL policy termination.

enrolled by the time DBTL was made mandatory, while about 20 percent enrolled afterward. The remaining 20 percent of household accounts had not enrolled in DBTL by the time of its termination in March 2014. Unsubsidized household gas remained available to households that did not enroll in DBTL either due to genuine exclusion (e.g., lack of a bank account or Aadhaar) or to those that voluntarily opted out of subsidies despite having bank accounts and Aadhaar. This also implied that unsubsidized household gas remained available to fake household gas accounts.

E. DBTL Rollout

The DBTL policy was introduced for voluntary enrollment in about half of the districts in India, staggered over six phases beginning in June 2013 (for the timeline, see Figure 3). Each phase consisted of a three-month voluntary enrollment period for households, to be followed by a mandatory required enrollment in DBTL for availing gas subsidies. By the time DBTL was repealed in March 2014, it had been mandatory in 18 districts (Phase 1 districts) for six months and another 22 districts (Phase 2 districts) for two months. More details on the policy rollout are provided in online Appendix A.

The selection of districts into early or late DBTL phases was unrelated to any observed pattern in household gas sales or diversion. The primary criterion for a district to be included in DBTL was Aadhaar coverage. The central government

considered voluntary DBTL enrollment only in those districts where Aadhaar coverage was likely to reach 80 percent soon, and DBTL became mandatory in these districts only when Aadhaar enrollment reached at least 80 percent of the population (Hota 2014). In June 2013, the average Aadhaar coverage in Phase 1 districts was more than 91 percent, and it approached 100 percent in these districts by January 2014.

F. DBTL Policy Termination and Reimplementation

On January 30, 2014, the Cabinet Committee on Political Affairs unexpectedly announced the decision to terminate the DBTL policy in the near future; the decision went into effect on March 10, 2014. The household gas subsidy transfer system was restored to its original structure prior to the DBTL rollout, and roughly 17 million households were reverted from bank transfer of the subsidy to traditional subsidy disbursement (Nambiar 2014). The government cited implementation-related issues as the main reason for DBTL's termination, though political lobbying by vested interests may have played an important role. Both the expected electoral costs of reduced access to corrupt funds and the amplification of concerns by special interest groups with vested interests seem to have contributed to its abrupt termination.⁷ The newly elected government restarted the DBTL policy in November 2014, covering all districts in two phases. Online Appendix A provides more details on DBTL repeal and reimplementation.

G. Additional Considerations Regarding DBTL Policy

There are a number of reasons why the policy reform may fail to reduce the diversion of household gas. New channels of household gas diversion may open up in response to higher expected profits in the black market. In the absence of rigorous validation of the household address, Aadhaar, and bank account, middlemen may collude with households to set up gas accounts for different members of the same household given the low fixed cost and high expected profit in the resale market. Genuine households may even consider independently reselling unused gas from their annual quota given the high expected profits.

II. Data

This study utilizes a combination of administrative and survey data. I use administrative data on gas refill sales by three government-controlled OMCs to estimate the impact of DBTL policy rollout and termination on gas sales. Data on black market prices, which I collected in a survey, is used to estimate the impact of DBTL's termination on black market gas prices. Online Appendix B provides a detailed description of the datasets used in this study. Data and code for this study are available through the American Economics Association repository (Barnwal 2024).

⁷Fuel subsidy reforms are generally susceptible to political business cycles in developing countries. For example, in Indonesia, "Parliamentary and presidential elections in 2014 mean that there is likely to be little appetite for further subsidy reforms in the first two-thirds of 2014" (IISD 2014).

TABLE 1—DESCRIPTIVE STATISTICS: ADMINISTRATIVE DATA

	Mean	Median	SD
<i>Panel A. Distributor level gas sales</i>			
14.2kg gas refills (household)	5,419	3,504	5,809
Subsidized	5,247	3,382	5,641
Unsubsidized	172.0	0	672.1
19kg gas refills (commercial)	255.2	40	784.8
Distributors		19,527	
Districts		681	
States		37	
Time period	34 months (Jan. 2013–Oct. 2015)		
<i>Panel B. Household level gas refills purchase (HPCL only)</i>			
Subsidized 14.2kg gas refills	6.523	7	2.853
Total 14.2kg gas refills	6.575	7	2.935
Monthly 14.2kg gas refills	0.553	1	0.586
Households		3.79 million	
Distributors		3,165	
Districts		509	
States		25	
Time period	12 months (April 2013–March 2014)		
Transactions		23.2 million	

Notes: Panel A shows distributor-level sales data for household and commercial gas from all three OMCs (BPCL, HPCL, and IOCL). Panel B shows details of household account-level gas refills (14kg refills) transactions data obtained from HPCL. This sample represents about 10 percent of HPCL household gas accounts (i.e., about 2.5 percent of all household beneficiaries) during the financial year 2013–2014.

A. Administrative Data on Cooking Gas Sales

Distributor-month level sales data includes all household gas refills from January 2013 to August 2015 (descriptive statistics are reported in Table 1). On average, a gas distributor sold 5,419 household refills in 14.2kg cylinders, which include both subsidized and unsubsidized household gas refills. While gas distributors sell only 172 unsubsidized household gas refills on average, there is a large variation across months and across distributors. The data also provide information on monthly sales of 19kg cylinder commercial gas refills since these are sold by the same distributors that sell household gas refills. These distributors sold 255 commercial gas refills per month on average. In the period before DBTL, total unsubsidized gas sales (combining commercial and unsubsidized household gas) comprised roughly 10 percent of total gas sales in the sample. Commercial gas sales (19kg cylinders) made up about 6 percent of total gas sales by distributors in the sample. In 2013–2014, total commercial, industrial, and transport gas consumption constituted about 12 percent of total gas consumed in India (Ministry of Petroleum and Natural Gas 2015). Commercial users can also purchase gas in bulk (e.g., 35 kg, 47.5 kg, and 425 kg commercial cylinders, or piped supply). These purchases primarily take place through exclusive commercial gas distributors which are not included in this study. Data on gas sales for transportation, sold through a separate supply chain, is also not included.

In addition, I use transaction-level information on 3.79 million household accounts to construct a household-month-level panel dataset from April 2013 to March 2014. This sample was obtained from a web portal maintained by Hindustan Petroleum Corporation Limited (HPCL), one of the three Oil Marketing Companies. Each web page on HPCL's Transparency Portal displayed 20 household accounts

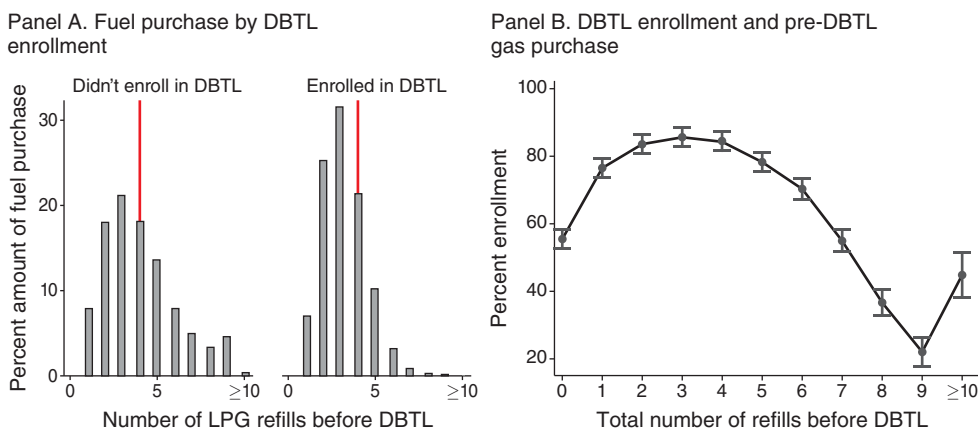


FIGURE 4. PRE-DBTL FUEL PURCHASE AND HOUSEHOLD ENROLLMENT IN DBTL

Notes: The above plot shows the distribution of pre-DBTL household gas refill (14.2kg) purchases by the DBTL enrollment status of household accounts in Phase 1 districts. The vertical axis in subplot A reports the percentage share of gas purchased by household accounts in the five-month period (April–August 2013) before DBTL policy became mandatory in Phase 1 districts. The vertical lines (in red) indicate the prorated annual LPG refill cap in 2013–2014, i.e., 3.75 gas refills rounded off to four LPG refills during five months period. The annual cap for LPG subsidy was nine 14.2 kg LPG refills in this time period, which explains the bunching at nine refills in subplot A. Subplot B shows the percentage of households enrolled in DBTL over the number of gas refills purchased during the pre-DBTL period in Phase 1 districts. The data used is collapsed to the distributor-by-refills level. Enrollment status of households is defined considering as of March 9, 2014, i.e., one day before the DBTL policy termination. Ninety-five percent confidence interval is shown. More than 10 LPG refills are combined together in both subplots.

in their entirety. Given the enormous amount of data on the portal, I used automated scripts to randomly select 10 percent of web pages for each distributor and obtained refill information on all households listed on those pages. The resulting sample of 23.2 million transactions in 509 districts from 25 states represents 10 percent of HPCL's about 40 million household gas accounts. The median household purchased seven subsidized gas refills (seven 14.2kg units $\simeq$ 100kg gas) in a year and no unsubsidized gas refill (Table 1). Transaction-level information also allows us to see heterogeneity in DBTL enrollment. For instance, household accounts purchasing larger amounts of gas before DBTL are less likely to enroll in DBTL (Figure 4).

B. Survey of the Black Market for Cooking Gas

I use black market price data collected in two survey rounds: around the time when DBTL was in force (January–February 2014) and after its termination (March 2014). Thus, the survey period overlaps with that of the administrative data. These surveys collected information from both gas deliverymen and small businesses. The objective of these surveys was to capture the DBTL rollout effect in Phase 3–6 districts, but the unexpected DBTL termination made this impossible. On the other hand, the termination of DBTL facilitated studying the impact of a sudden policy change in districts where DBTL was in place.

The audit design of the fieldwork leverages the fact that deliverymen, who are tasked with doorstep supply of both household and commercial fuel, become the

TABLE 2—DESCRIPTIVE STATISTICS: AUDIT SURVEY IN THE BLACK MARKET

Variable	Observations	Mean	SD	Min	Max
Small business survey price (₹)	2,369	1,039.13	241.26	430	1,600
Deliverymen survey price (₹)	1,202	1,062.49	233.41	550	1,950
Small businesses			1,452		
Deliverymen			1,202		
Districts			89		
States			11		

Notes: The above table shows summary statistics of black market prices for 14.2 kg gas refills collected from the deliverymen and small businesses in two rounds of survey (before and after the policy termination). Price values are in ₹ (2013–2014 exchange rate: US\$ 1 = ₹60 approximately). Price values reported by the respondents, which are below the average OMC price for subsidized fuel or above the average OMC price for nonsubsidized commercial fuel, are dropped. The small-business survey had a 20 percent attrition rate in the post-termination round.

de facto suppliers in the black market. The survey team selected up to seven deliverymen per district in urban market areas in 89 districts. This is complemented with a survey of 15 small businesses in the same urban markets. This small-business survey collected information on current black market prices. The small business survey also elicited retrospective data on the date and price of their last five gas refills. This refill history information provides an unbalanced daily panel of black market prices and allows testing for any discontinuity in black market prices due to policy termination. The deliverymen survey is not a panel since it was not feasible to follow the same deliverymen over time, but the small-business survey is a panel. The average price of a black market gas refill was ₹1,039 as reported by small businesses, while it was ₹1,062 when elicited from deliverymen. This small difference in prices can be potentially explained by longstanding supplier-buyer relationships in the black market. Table 2 provides descriptive statistics on these surveys. More details on the survey implementation are provided in online Appendix B.

C. Additional Datasets

The National Sample Surveys (NSS) provide information on household gas purchases. Two rounds of NSS (2009–2010 and 2011–2012) are used to test for parallel trends in household gas usage across DBTL and non-DBTL districts (NSSO 2010, 2012).

I further use data on household-level total expenditure and cooking fuel expenditure from the Center for Monitoring of Indian Economy (CMIE) surveys to estimate the impact of DBTL termination on the household cooking fuel expenditure share (Vyas 2017). CMIE surveys are available starting from January 2014, which limits the data analysis to DBTL policy termination only. The survey question on monthly expenditure asks, “What was the expenditure incurred by the household as a whole on cooking fuel?” This includes expenditures on all types of fuel used for cooking purposes (gas, kerosene, wood, or coal), but not electricity.⁸ On average, cooking fuel accounts for

⁸Households under DBTL can potentially interpret gas prices in two ways (with or without subsidy) when responding to the CMIE survey. Since the transfer under DBTL is tied to gas purchase, the former is the correct way

about 4.8 percent of household monthly expenditures. The CMIE data provides an independent measure of actual cooking fuel expenditure by legitimate households in a population-representative sample. Household gas diverted to ghost accounts should not show up in this response since this is a household survey, and it is reasonable to assume that legitimate households report only their own cooking fuel expenditures.

III. Empirical Strategy

I exploit the staggered rollout of DBTL as well as its unexpected termination in early 2014 to estimate the impact of DBTL using a difference-in-differences identification strategy. The staggered rollout provides policy treatment variation across time and districts. Post-termination, the ex ante status quo in the gas subsidy system was restored. Thus, this setting allows me to estimate the impact of the DBTL policy as well as its termination.

Equation (1) shows the main empirical specification used with the distributor-month level panel data:

$$(1) \quad Y_{idsm} = \alpha + \beta Post_m \times DBTL_d + \mu_i + \delta_m + \gamma_{sm} + \epsilon_{idsm}.$$

The outcome variable Y_{idsm} denotes $\log(gas\ sales)$ for distributor i in district d , state s , and month m . The same specification is used for different types of gas sales. Distributor fixed effects μ_i are included to control for distributor-level time-invariant factors; δ_m denotes month fixed effects; and γ_{sm} denotes a state-month linear trend. Alternatively, state-by-month fixed effects are used in place of month fixed effects and the state-month linear trend. State-specific time factors control for seasonality and any state-level policy changes over time. District fixed effects are not included since they are subsumed by distributor fixed effects. Standard errors are clustered at the district level. To estimate the impact of DBTL rollout, data covering the period before DBTL and during DBTL are used. The difference-in-differences coefficient β provides the average treatment effect of the DBTL policy on fuel sales. In the preferred specification, DBTL Phase 1 districts comprise the policy treatment group. The pooled control group consists of districts assigned to later phases (i.e., Phases 3–6) as well as nonpolicy districts where the DBTL policy was not yet scheduled for implementation. Therefore, the policy treatment period covers six months of refill sales.⁹ The effect of policy termination is estimated by replacing $Post_m$ with $Post-termination_m$ and using data from the time period starting after DBTL rollout.

I further estimate an event study model with month dummies interacted with policy treatment assignment (equation (2)):

$$(2) \quad Y_{idsm} = \sum_{\tau=Jan2013}^{Oct2014} \beta_{\tau} DBTL_d \times \mathbf{1}\{m = \tau\} + \mu_i + \delta_m + \gamma_{sm} + \epsilon_{idsm},$$

to answer this question. If a significant proportion of households interpret this question differently and report their cooking fuel expenditure considering gas to be unsubsidized while receiving the subsidy in their bank account, it will be more difficult to reject the null hypothesis of no increase in cooking fuel expenditure share due to DBTL.

⁹The impact on gas sales in Phase 2 districts, where DBTL policy was mandatory for a relatively short period of only two months, is estimated separately.

where β_τ provides the month-specific policy effect on gas sales.

I use a similar specification for the account-level monthly panel data constructed using transaction-level information on gas purchases by household accounts. Regressions using the household-month panel include household account and month fixed effects. Household-month-level data allows additional tests to study household-level diversion, heterogeneous impacts, and delayed household DBTL enrollment.

Finally, I estimate the impact of DBTL policy termination on black market fuel prices using price survey data and a specification similar to equation (1). Here, the data include one round before and one round after the policy termination. In the preferred specification, the outcome variable is $\log(\text{black market price})$. The black market price data consists of individual price observations collected in the survey of deliverymen or small businesses. The small business survey data analysis includes small business fixed effects. Because I do not observe the same deliverymen over time, I instead use district fixed effects.

Online Appendix C presents a framework to guide the discussion of the empirical results described in the next section. Using a simple model of supply and demand of gas diverted to the black market, I derive comparative statistics to understand the impact of removing dual pricing on three primary outcomes of interest: fuel sales to household accounts, fuel sales to commercial accounts, and prices in the black market.

Parallel Trends and Placebo Tests

Identification in this empirical design relies on the parallel trend assumption, that is, DBTL and non-DBTL districts following similar trends in gas sales and household gas usage before DBTL was introduced. I test for parallel pre-trends (i.e., trends prior to DBTL implementation) using both administrative data on gas sales to household and commercial accounts and with a placebo test using National Sample Survey (NSS) data on household gas usage.

Using administrative data on gas sales in the pre-DBTL period (January–August 2013), columns 1 and 2 of Table 3 show the estimated monthly difference in gas sales to household accounts across treatment and control districts. Columns 3 and 4 show the same for commercial gas sales. The specification used here is similar to equation (2) with state and month fixed effects in even-numbered columns and state-by-month fixed effects in odd-numbered columns. The lack of differential trends in the pre-period supports the parallel trends assumption. I cannot reject the joint equality of the estimated coefficients in all cases except in one commercial gas specification showing a p -value of 0.0736 (column 4). When compared to the main results shown in the next section, the magnitude of these fairly precise estimates in column 4 is relatively small.

Next, I use a placebo policy approach to rule out the possibility that persistent unobserved factors in genuine households' fuel consumption are driving the main result on household gas. The estimated difference-in-differences coefficient in the placebo test conducted using 2009–2010 and 2011–2012 NSS data is close to zero in magnitude and statistically insignificant (online Appendix Table D.1). This confirms that household-level gas usage was on a similar path across DBTL and non-DBTL districts prior to the policy's introduction. This is consistent with the fact that the

TABLE 3—PRE-TRENDS IN GAS SALES

	log(<i>Household LPG sales</i>)		log(<i>Commercial LPG sales</i>)	
	(1)	(2)	(3)	(4)
<i>DBTL × Jan.–Feb. 2013</i>	0.0490 (0.0315)		0.0595 (0.0466)	
<i>DBTL × Mar.–Apr. 2013</i>	0.0343 (0.0318)		−0.0136 (0.0494)	
<i>DBTL × May–June 2013</i>	0.000899 (0.0163)		0.00848 (0.0419)	
<i>DBTL × Jan. 2013</i>		0.0433 (0.0311)		0.0672 (0.0590)
<i>DBTL × Feb. 2013</i>		0.0462 (0.0479)		0.0383 (0.0716)
<i>DBTL × Mar. 2013</i>		0.0450 (0.0437)		−0.00904 (0.0625)
<i>DBTL × Apr. 2013</i>		0.0152 (0.0345)		−0.0310 (0.0604)
<i>DBTL × May 2013</i>		0.0243 (0.0297)		0.0217 (0.0635)
<i>DBTL × June 2013</i>		−0.0313 (0.0179)		−0.0165 (0.0622)
<i>DBTL × July 2013</i>		−0.00869 (0.0192)		−0.0132 (0.0685)
<i>Constant</i>	8.324 (0.00441)	8.258 (0.0372)	4.884 (0.00718)	4.999 (0.0775)
Observations	87,079	87,079	70,686	70,686
R^2	0.955	0.956	0.935	0.936
Distributor FE	Yes	Yes	Yes	Yes
State-month or State-bimonth FE	Yes	Yes	Yes	Yes
Pr(> F)	0.453	0.165	0.228	0.0736

Notes: This table shows the difference in gas refill purchases between DBTL and non-DBTL districts over time. Both 14.2kg household gas refills (columns 1 and 2) and 19kg commercial gas refills (columns 3 and 4) are shown. The treated group consists of DBTL Phase 1 districts where the policy was enforced in September 2013. July–August 2013 bin is set as the base category. Districts in Phases 3–6 and districts not on the DBTL implementation schedule comprise the control group. Data is at the distributor-month level. Pr(> F) shows the p -value for F -test of equivalency of coefficients. Standard errors, clustered at the district level, are in parentheses.

selection of districts in DBTL phases was on the basis of Aadhaar coverage, and not based on any preexisting trends in fuel usage or diversion.

IV. Results

A. Impact of DBTL Policy on Gas Purchased by Household Accounts

Results for the impact of DBTL rollout on gas sales to household accounts are reported in Table 4. Columns 1 and 2 indicate 17 percent and 19 percent reductions in household gas sales, respectively.¹⁰ Estimates in even-numbered columns include

¹⁰Since the outcome variable is log-transformed, $\beta \times 100\%$ estimates reported here are good approximations of the “true” effects $(e^\beta - 1) \times 100\%$. However, for large values of β , the true effect will be much larger. I report $e^\beta - 1$ along with β in such cases.

TABLE 4—IMPACT OF DBTL ON HOUSEHOLD GAS SALES

	$\log(\text{Household LPG sales})$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>DBTL</i> $\times$ <i>post</i>	−0.167 (0.0297)	−0.188 (0.0341)	−0.174 (0.0304)	−0.189 (0.0349)	−0.113 (0.0246)	−0.154 (0.0519)
Observations	160,879	160,879	90,900	90,900	76,166	76,166
R^2	0.942	0.943	0.946	0.947	0.938	0.939
Distributor FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE, state-month trend	Yes		Yes		Yes	
State-month FE		Yes		Yes		Yes
Control districts	Phase 3–6 and nonpolicy		Phase 3–6		Nonpolicy	

Notes: This table shows difference-in-differences estimates of the impact of DBTL policy on gas sold to household accounts. The data consists of distributor-month level panel covering the time period before and during the policy reform (January 2013 to February 2014). The outcome variable is the log of total household gas refills sold in a given month. The DBTL policy group consists of DBTL Phase 1 districts. Columns 1 and 2 combine districts in Phase 3–6 and districts not on the DBTL implementation schedule by March 2014 together as the control group; columns 3 and 4 present estimates for a control group including districts in Phase 3–6 only; and columns 5 and 6 for a control group including only those districts not on the DBTL implementation schedule. Phase 2 districts are not included. All estimates include distributor fixed effects and either month fixed effects and state-month trends or state-month fixed effects. Standard errors, clustered at the district level, are in parentheses.

state-month fixed effects and are preferred. The first two columns include all districts with voluntary DBTL or no DBTL policy in the control group. The next four columns provide robustness checks using different control groups: columns 3 and 4 show estimates with DBTL Phases 3 to 6 (i.e., districts with voluntary DBTL enrollment and scheduled for mandatory DBTL) as the control group, while columns 5 and 6 use nonpolicy districts (i.e., districts not yet scheduled for DBTL) as the control group. Figure 5 shows the regression-adjusted monthly means of household gas sales and demonstrates the reduced gas purchases in DBTL districts compared with control districts. Phase 2 districts also show similar results (reported in online Appendix Figure D.4 and Table D.2).¹¹ The estimated effect in the first month is higher in magnitude, likely due to inter-temporal substitution of the purchase of gas refills by households waiting to enroll in DBTL.

The DBTL policy termination in March 2014 provides another opportunity to study how the policy reform affected gas sales. Figure 5 shows the impact of policy termination on household gas sales for DBTL districts in their reversion to levels seen in the control districts. Columns 1 and 2 of Table 5 report the effect as a 14 percent increase in gas purchase by household accounts following the reversal of the DBTL policy. Taken together, the results suggest symmetric responses to policy implementation and policy termination.

The results discussed above combine both types of household gas refills, unsubsidized and subsidized. Online Appendix Table D.3 (columns 1–4, panels A

¹¹ These results for Phase 2 districts are similar to those estimated in the case of Phase 1 districts, though magnitudes are higher for household gas sales. DBTL Phase 2 districts observed a mandatory DBTL period of only two months (January and February 2014), so factors such as anticipation, storage, and strategic delays by users in purchasing gas refills once the political announcement to terminate the policy was made, are more likely to affect gas refills sales.

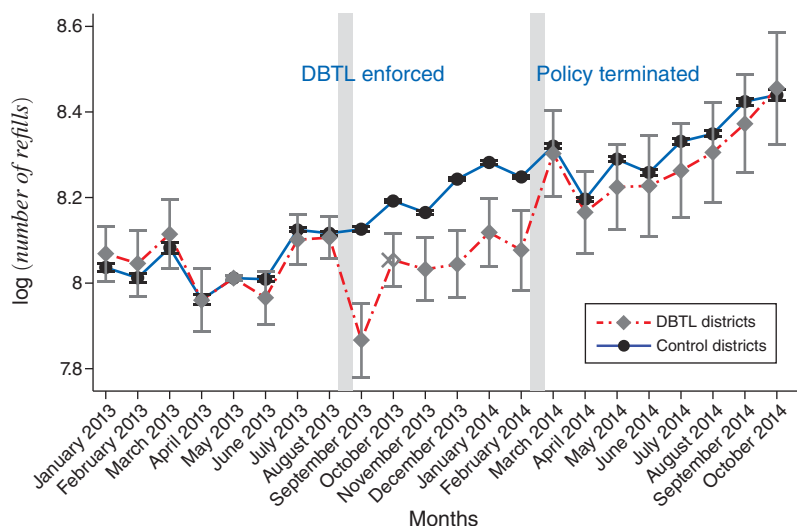


FIGURE 5. IMPACT OF DBTL ON HOUSEHOLD GAS SALES BY DISTRIBUTORS

Notes: The above plot shows the impact of DBTL on household gas sales using a distributor-month panel dataset. The outcome variable is the log of the total number of 14.2kg household gas refills sold. This plot reports month-wise regression-adjusted means of household fuel sales in DBTL policy and control districts. Distributor and state-by-month fixed effects are included. DBTL policy districts include all districts where DBTL was enforced in Phase 1 in September 2013, whereas the control group includes districts with voluntary DBTL as well as those where DBTL was not yet scheduled for implementation. Phase 2 districts and two districts in Phase 1 where DBTL was enforced after September 2013 are excluded. As shown by the vertical bars, DBTL was enforced on September 1, 2013, and terminated on March 10, 2014. Ninety-five percent confidence intervals shown are for the regression-adjusted mean sales by distributors in the DBTL policy and control groups. Standard errors are clustered at the district level.

TABLE 5—IMPACT OF DBTL TERMINATION ON HOUSEHOLD GAS SALES

	log(<i>Household LPG sales</i>)					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>DBTL × post termination</i>	0.136 (0.0241)	0.141 (0.0261)	0.141 (0.0242)	0.143 (0.0262)	0.126 (0.0236)	0.187 (0.0382)
Observations	179,404	179,404	100,422	100,422	85,571	85,571
R^2	0.943	0.944	0.949	0.950	0.937	0.938
Distributor FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE, state-month trend	Yes		Yes		Yes	
State-month FE		Yes		Yes		Yes
Control districts	Phase 3–6 and nonpolicy		Phase 3–6		Nonpolicy	

Notes: This regression shows difference-in-differences estimates for the DBTL policy termination on household gas sales. Data consists of distributor-month level panel data during the policy reform and after its termination (September 2013 to October 2014). The outcome variable is the log of total household gas refills sold to households in a given month. The DBTL policy group consists of DBTL Phase 1 districts. Columns 1 and 2 combine districts in Phase 3–6 and districts not on the DBTL implementation schedule by March 2014 together as the control group; Columns 3 and 4 present estimates for a control group including districts in Phase 3–6 only; and columns 5 and 6 for a control group including only those districts not on the DBTL implementation schedule. Phase 2 districts are not included. All estimates include distributor fixed effects and either month fixed effects and state-month trends or state-month fixed effects. Standard errors, clustered at the district level, are in parentheses.

TABLE 6—IMPACT OF DBTL ON COMMERCIAL GAS SALES

	<i>log(Commercial LPG sales)</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>DBTL × post</i>	0.126 (0.0370)	0.136 (0.0421)	0.115 (0.0373)	0.133 (0.0422)	0.130 (0.0455)	0.0298 (0.0811)
Observations	129,668	129,668	76,903	76,903	57,933	57,933
<i>R</i> ²	0.920	0.923	0.929	0.933	0.904	0.907
Distributor FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE, state-month trend	Yes		Yes		Yes	
State-month FE		Yes		Yes		Yes
Control districts	Phase 3–6 and nonpolicy		Phase 3–6		Nonpolicy	

Notes: This regression shows difference-in-differences estimates for the impact of the DBTL policy on commercial gas sales. Data consists of distributor-month level panel data before and during the policy reform (January 2013 to February 2014). The outcome variable is the log of the total 19kg commercial gas refills sold to commercial accounts in a given month. The DBTL policy group consists of DBTL Phase 1 districts (16 districts). Columns 1 and 2 combine districts in Phase 3–6 and districts not on the DBTL implementation schedule by March 2014 together as the control group; columns 3 and 4 present estimates for a control group including districts in Phase 3–6 only; and columns 5 and 6 for a control group including only those districts not on the DBTL implementation schedule. Phase 2 districts are not included. All estimates include distributor fixed effects and either month fixed effects and state-month trends or state-month fixed effects. Standard errors, clustered at the district level, are in parentheses.

and B) further breaks down household gas sales into subsidized and unsubsidized categories. The impact of DBTL rollout and termination on subsidized household gas sales is much higher than the impact on total household gas sales shown in Table 4. In contrast to subsidized gas sales, unsubsidized gas sales to household accounts increased (decreased) significantly with the DBTL rollout (termination). Specifically, the policy change reduced subsidized household gas sales by 46 percent ($= e^{0.38} - 1$) and increased unsubsidized household gas sales by tenfold ($= e^{2.37} - 1$), though from a much smaller baseline level. This makes sense from the diversion standpoint since unsubsidized household gas, available to all household accounts irrespective of DBTL enrollment, was still cheaper than commercial gas (Figure 2, panel A), and hence it remained prone to diversion to the black market.

B. Market Effects of DBTL Policy

Impact on Gas Purchased by Commercial Accounts.—Commercial gas purchases went up by 13 percent to 14 percent in response to the DBTL policy (columns 1 and 2, Table 6). Further, the DBTL policy termination lowered commercial gas sales by 8 percent to 12 percent (columns 1 and 2, Table 7). These estimates are overall robust to the use of different control groups (columns 3–6, both tables), except in two out of twelve specifications where the coefficients have the correct sign but are less precisely estimated.¹² Figure 6 reports the regression-adjusted monthly means of commercial gas purchases in DBTL and non-DBTL districts. Commercial users

¹²One-third of the states where DBTL Phase 1 districts come from are not in the second control group here (i.e., districts where DBTL was not yet scheduled for implementation), which may limit power in specifications that include state-specific time effects.

TABLE 7—IMPACT OF DBTL POLICY TERMINATION ON COMMERCIAL GAS SALES

	$\log(\text{Commercial LPG sales})$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>DBTL $\times$ post termination</i>	−0.0753 (0.0377)	−0.132 (0.0437)	−0.0660 (0.0387)	−0.136 (0.0439)	−0.0505 (0.0392)	−0.113 (0.0600)
Observations	141,174	141,174	83,533	83,533	63,174	63,174
R^2	0.917	0.920	0.927	0.930	0.901	0.903
Distributor FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE, state-month trend	Yes		Yes		Yes	
State-month FE		Yes		Yes		Yes
Control districts	Phase 3–6 and nonpolicy		Phase 3–6		Nonpolicy	

Notes: This regression shows difference-in-differences estimates for the impact of the DBTL policy termination on commercial LPG sales. Data consists of distributor-month level panel data during the policy reform and after its termination (September 2013 to October 2014). The outcome variable is the log of the total 19kg commercial LPG refills sold to commercial accounts in a given month. The DBTL policy group consists of DBTL Phase 1 districts (16 districts). Columns 1 and 2 combine districts in Phase 3–6 and districts not on the DBTL implementation schedule by March 2014 together as the control group; columns 3 and 4 present estimates for a control group including districts in Phase 3–6 only; and columns 5 and 6 for a control group including only those districts not on the DBTL implementation schedule. Phase 2 districts are not included. All estimates include distributor fixed effects and either month fixed effects and state-month trends or state-month fixed effects. Standard errors, clustered at the district level, are in parentheses.

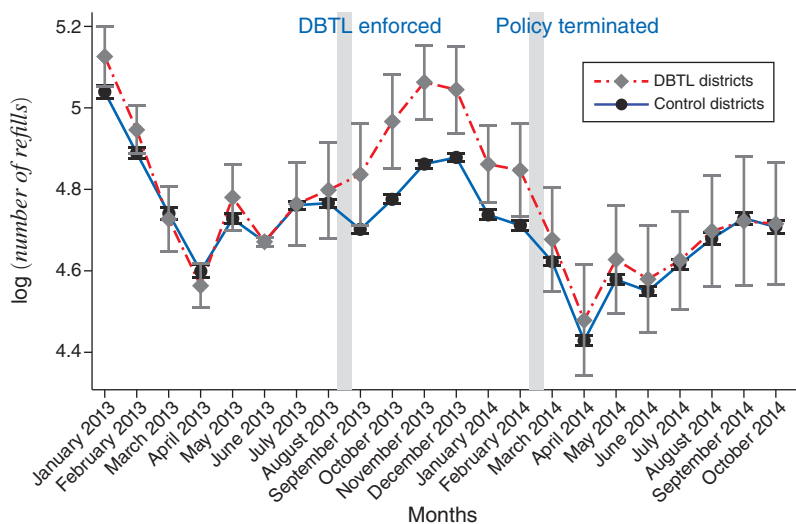


FIGURE 6. IMPACT OF DBTL ON COMMERCIAL GAS SALES BY DISTRIBUTORS

Notes: The above plot shows the impact of DBTL on commercial gas sales using a distributor-month panel dataset. The outcome variable is the log of the total number of 19kg commercial gas refills sold. This plot reports month-wise regression-adjusted means of commercial fuel sales in DBTL policy and control districts. Distributor and state-by-month fixed effects are included. DBTL policy districts include all districts where DBTL was enforced in Phase 1 in September 2013, whereas the control group includes districts with voluntary DBTL as well as those where DBTL was not yet scheduled for implementation. Phase 2 districts and two districts in Phase 1 where DBTL was enforced after September 2013 are excluded. As shown by the vertical bars, DBTL was enforced on September 1, 2013, and terminated on March 10, 2014. Ninety-five percent confidence intervals shown are for the regression-adjusted mean sales by distributors in the DBTL policy and control groups. Standard errors are clustered at the district level.

responded to the DBTL policy by purchasing more 19kg commercial gas refills, even though the policy change did not affect commercial gas pricing directly.

Commercial users could also buy unsubsidized household gas refills in the black market after the policy change. Combining both commercial and unsubsidized household gas sales, online Appendix Table D.3 (columns 7 and 8, panel A) shows that total unsubsidized gas sales by gas distributors increased by 263 percent ($= e^{1.29} - 1$) in response to DBTL. One concern remains that not all the increase in unsubsidized gas purchases can be attributed to diversion since at least a part of it may be by genuine households due to voluntary or involuntary nonenrollment in DBTL. I discuss this issue further in Section IVD and provide evidence supporting that most of the increase in unsubsidized household gas purchases was likely due to diversion. Nevertheless, since the data does not allow disentangling these two components in a clear way, Tables 4 to 7 can be interpreted as providing a conservative estimate of DBTL's impact on diversion.

In the absence of household gas diversion and subsequent reselling through the black market, DBTL should not have had any impact on commercial gas sales. The observed effect of this policy reform on the black market may help explain this response.

Impact on Black Market Prices.—As noted earlier, the survey data does not cover the period before DBTL implementation in Phase 1 and 2 districts, so I can only estimate the impact of DBTL termination. Table 8 (panel A) shows that DBTL policy termination reduced black market price quotes by deliverymen by 13 percent (panel A, column 1). After controlling for survey date and state-by-survey-round fixed effects, columns 2 and 3 show a reduction of 16 percent and 18 percent, respectively (columns 2 and 3). Price data collected from small businesses show an impact of similar magnitude (panel B). Here, nonpolicy districts comprise the control group due to concerns about anticipation effects on prices in districts with voluntary DBTL. In this analysis, small business and location fixed effects help address any consistent misreporting across survey rounds, while price reports collected simultaneously from two sides of the market (i.e., deliverymen and small businesses) help in ruling out bias in estimates due to strategic misreporting. Online Appendix Table D.6 provides robustness checks using different sets of districts in the control group. These results confirm that the policy termination unambiguously brought black market gas prices down.

The small business survey also elicited retrospective data on the date and price of the last five gas refills. This refill history information provides an unbalanced daily panel of black market prices and allows testing for any discontinuity in black market prices due to policy termination. Figure 7 shows the results using a 40-day window around the policy termination date. Note that the discontinuity in black market prices only appears in the DBTL districts, whereas price levels in the voluntary DBTL and nonpolicy districts are stable around the policy termination date. Online Appendix Table D.7 provides additional results using the price data from refill history information.

Taken together, the results on black market prices and on commercial and household gas sales indicate that restoring the household gas dual-pricing regime increased the diversion of subsidized gas to the black market, causing a positive

TABLE 8—IMPACT OF DBTL POLICY TERMINATION ON BLACK MARKET GAS PRICES

	$\log(\text{price})$					
	Panel A. Deliverymen survey			Panel B. Small Business survey		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>DBTL</i> $\times$ <i>post termination</i>	−0.127 (0.0401)	−0.159 (0.0389)	−0.183 (0.0293)	−0.186 (0.0617)	−0.189 (0.0612)	−0.198 (0.0485)
Observations	504	504	504	796	796	796
R^2	0.756	0.778	0.862	0.744	0.771	0.783
District FE	Yes	Yes	Yes			
Small business FE				Yes	Yes	Yes
Survey date FE		Yes	Yes		Yes	Yes
State by survey round FE			Yes			Yes

Notes: This table presents difference-in-differences estimates of the impact of DBTL termination on black market gas refill prices. Two rounds of surveys (during DBTL and after the policy termination) collected information on gas prices from gas deliverymen in an audit survey and from small businesses in interviews. The outcome variable, i.e., the price for a household gas refill sold in the black market (₹), is log-transformed. Results are shown separately using price information collected from deliverymen in panel A and from small businesses in panel B. The treatment group consists of the DBTL districts (Phase 1 and 2) and the control group of districts where DBTL was not yet scheduled for implementation, totaling 38 districts in 8 states. District fixed effects are included in panel A and small business (firm) fixed effects are included in panel B. Survey date fixed effects are used in columns 2 and 4. State-by-survey round fixed effects are included in columns 3 and 6. Standard errors, clustered at the district level, are in parentheses.

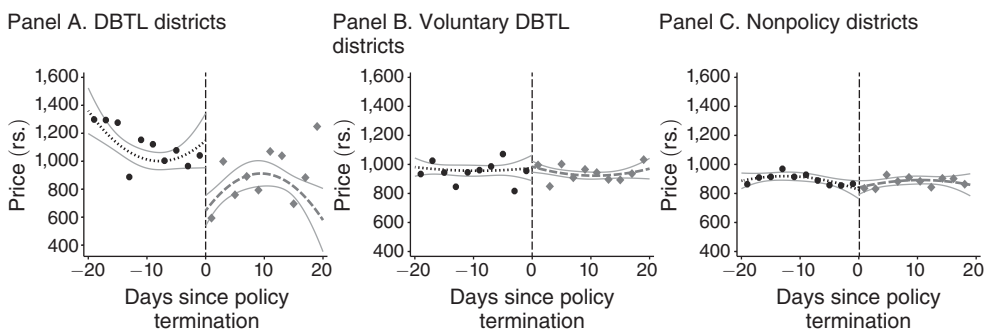


FIGURE 7. POLICY TERMINATION: IMPACT ON PRICES IN THE GAS BLACK MARKET

Notes: This plot shows the discontinuity in black market LPG refill prices at the time of DBTL termination in DBTL districts (panel A). The price and refill dates reported by commercial users in the survey are used to construct the time series of black market prices. A 20-day window is used on either side of the DBTL termination date. A fitted quadratic curve is shown with a 95 percent confidence interval. Panel A shows results using data from DBTL Phase 1 and 2 districts, while panels B and C show results using data from control districts (DBTL Phase 3–6 and nonpolicy districts). Nonpolicy districts include all districts where DBTL was not yet scheduled for implementation. While the government decided to terminate the policy on January 30, 2014, the official notification of DBTL termination was released on March 7, 2014 (Friday), and the policy was effectively terminated on March 10, 2014 (Monday). Thus, the exact DBTL policy termination date was not publicly known until the business day before the actual termination.

supply shock. This offers a reasonable explanation for the decline in the equilibrium black market price and the decline in gas purchases by commercial accounts as businesses turned more to the black market for gas. The discussion in online Appendix C provides a framework to further explain the potential mechanisms behind these results.

In this setting, it is important to understand how changes in household and commercial gas purchases are related quantitatively. For each kilogram decline in subsidized household gas purchases due to DBTL, unsubsidized household gas purchase rose 0.68 kg while commercial purchases (of 19 kg units) rose 0.05 kg. This suggests that leakage reduction occurred in two ways: first, through a large decline in subsidies provided to household accounts, and second, through a smaller decrease in tax evasion by commercial users. Thus, while the DBTL policy likely reduced the subsidy leakage significantly, there remains a substantial tax-related leakage that DBTL was not designed to address. There are a few caveats here. First, an unknown amount of commercial purchases (when sold in units other than 19 kg) is missing in my data. Second, an overall decline in the amount of gas purchased by commercial users is expected when they face a higher price. Further discussion is provided in online Appendix D.

C. Service Quality: Delay in Delivery of Household Gas Refills

DBTL may have unintended impacts on service quality for household gas users. If DBTL reduces the rents for distributors and deliverymen, these agents may respond by reducing their effort in servicing households' gas refill requests in a timely manner. At the same time, by reducing the workload of these agents due to a decrease in household gas refills sales volume or curtailing their illicit (albeit time-consuming) outside options, DBTL may have improved the quality of service provided by distributors and deliverymen to households. The administrative data allows measuring one aspect of the service quality, namely, the time households have to wait for gas refills after ordering for one.¹³ Using administrative data on gas refill orders and delivery dates, I calculate household wait times for gas refills. On average, a household waits 3.2 days for a gas refill.

Figure 8 shows that, compared to the control group, the waiting period for gas refills was higher in DBTL districts during the first two months of DBTL, but it was consistently lower in the next four months during DBTL. One possible explanation is that a household that hadn't yet enrolled in DBTL to receive the subsidy in its bank account, may have delayed receiving a gas refill until DBTL enrollment. The estimated difference-in-differences coefficient (online Appendix Table D.8 shows that DBTL reduced the wait time by 0.79 days: that is, about 23 percent. These results suggest that gas distributors and deliverymen may have prioritized gas diversion before DBTL. Under DBTL, with the arbitrage opportunity provided by dual pricing weakened, the diversion of gas refills was reduced, and service quality to households observed an improvement.

D. Exclusion of Households in DBTL

There are potential alternative explanations for the estimated impact of DBTL on household gas sales other than reduced leakage, including voluntary nonenrollment (i.e., genuine households choosing not to enroll in DBTL); late enrollment (i.e.,

¹³ Households are required to pre-order gas refills via phone, web portal, or personal visit. The distributors primarily provide doorstep delivery of gas refills, although the wait time from order to delivery can vary significantly.

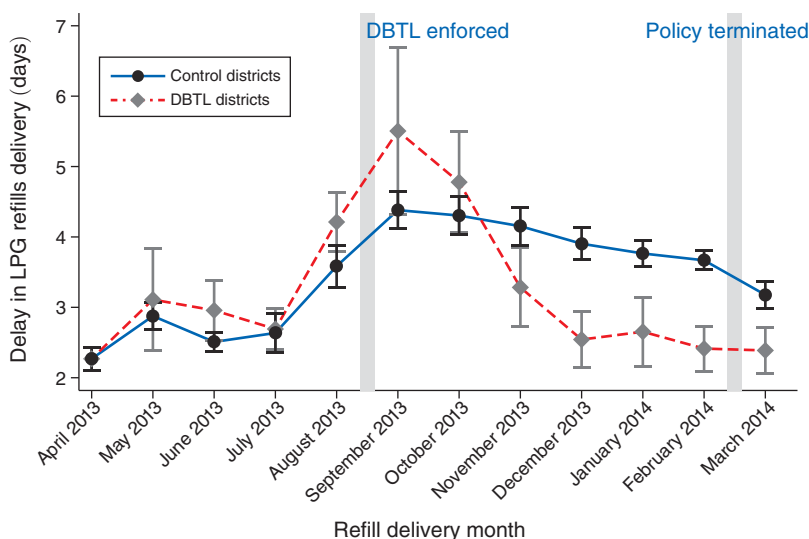


FIGURE 8. DBTL'S IMPACT ON PDS SERVICE DELIVERY

Notes: This plot shows the impact of DBTL policy on service quality observed by households. The outcome variable is the number of days taken by the distributor and deliverymen in delivering the gas refill, once a household has ordered it. This plot reports monthly regression-adjusted means of delay to household accounts in DBTL policy and control districts. Household and month fixed effects are included. Ninety-five percent confidence intervals shown are for the regression-adjusted mean delays by distributors in the DBTL policy and control groups. Standard errors are clustered at the district level.

genuine households enrolling in DBTL with a delay); or exclusion (i.e., genuine households unable to enroll in DBTL due to lack of a bank account or Aadhaar number). About 20 percent of household gas accounts in Phase 1 districts enrolled in DBTL after September 2013, and a similar proportion did not enroll at all during the six-month DBTL enforcement period. Households may delay enrolling until they need to purchase a refill. With the median household account buying seven gas refills in a year, many households can go on for months without a gas refill. Fake household accounts can not enroll in DBTL because of Aadhaar and bank account requirements. Assuming household accounts buying a higher number of subsidized refills in the pre-DBTL period are more likely to be fake, one would expect to see a lower proportion of such accounts enrolling in DBTL. Indeed, Figure 4 shows that household gas accounts with a higher take-up of gas subsidy in the pre-DBTL period are less likely to enroll in DBTL.

By January 2014, 98.5 percent of the population in Phase 1 districts had an Aadhaar (online Appendix Figure D.1). The 2011 census shows that the proportion of households with access to banking is higher than the proportion of households using gas (online Appendix Figure D.2). Average banking access was about twice the average gas usage in the 2011 Census, with a smaller, but still positive, difference in urban areas. Furthermore, gas users are predominantly from middle- to higher-income groups (Figure 1). While the core concerns (limited bank account ownership and Aadhaar coverage) are unlikely to be relevant for most gas consumer

households, they could have still caused the exclusion of some genuine household accounts.¹⁴

It would be particularly concerning if households had to increase their cooking fuel expenditures because of subsidy exclusions.¹⁵ Using household-month survey data from the Centre for Monitoring Indian Economy (CMIE), I estimate the impact of DBTL policy on households' cooking fuel expenditure share. Table 9 (column 1) shows that the estimated difference-in-differences coefficient is 0.0877 percent, though it is statistically insignificant and small in magnitude when compared with the average household monthly expenditure of 4.8 percent on cooking fuel. Including state-specific time controls leads to a similar conclusion (columns 2 and 3). The upper bound of the 95 percent confidence interval for the change in monthly expenditure on cooking fuel (column 3) suggests that there is only one chance in forty that the DBTL increased household cooking fuel expenditure share by more than 0.34 percentage points. Given the much larger gap between subsidized and unsubsidized household gas prices, DBTL policy on average does not seem to have had a statistically or economically significant impact on the share of household monthly expenditures spent on cooking fuel.

Even in the case of some exclusion, the resulting welfare losses may be outweighed by gains from the reduction in leakage. For three of four major commodities in the Food PDS (known as "ration shops") in India, Gadenne (2018) estimates that a leakage of more than 6 percent outweighs the welfare gains from redistribution and protection against price risk. For the gas public distribution system, she finds exceptionally high negative welfare gains before even considering any leakage. This is not surprising given the inequality in the incidence of gas subsidies in India as shown in Figure 1.

E. Distributional Effects of DBTL Policy

It would be a concern if the DBTL policy had adverse distributional implications despite the leakage reduction. To study this, I first show how the cooking fuel expenditure share changes with DBTL across expenditure groups. I then analyze the incidence of policy reform by the pre-DBTL level of gas subsidy received by household accounts.

Impact across Household Expenditure Quintiles.—The estimated average effect on cooking fuel expenditure, shown in Table 9, may mask heterogeneity across income groups. Particularly, if middle- and lower-income households¹⁶ are more vulnerable to exclusion from subsidies during DBTL (for instance, due to concerns related to bank accounts or Aadhaar access), enforcing the DBTL policy may lead

¹⁴Independently administered household surveys conducted in May 2015 after DBTL reimplementation indicate an enrollment rate of 94.6 percent (Jain, Agrawal, and Ganeshan 2018). Of the remaining 5.4 percent, about one-half of the households cite a lack of interest in gas subsidies, confirming self-selection out of the program. Only 0.2 percent of households cite a lack of a bank account as a reason for not enrolling in DBTL.

¹⁵Muralidharan, Niehaus, and Sukhtankar (2023) show that, along with leakage reduction, exclusion may also arise due to poor management of policy transition in welfare delivery programs, even when the structural features of the policy reform do not cause it.

¹⁶As mentioned in Section IB, low-income households were relatively less likely to be using gas as the primary cooking fuel, and so most of the gas subsidy was being availed by middle- to higher-income households.

TABLE 9—IMPACT OF DBTL ON HOUSEHOLD FUEL EXPENDITURE

	Cooking fuel's expenditure share		
	(1)	(2)	(3)
<i>DBTL × pre termination</i>	0.000877 (0.00112)	0.00154 (0.00121)	−0.000364 (0.00156)
Observations	1,296,514	1,296,514	1,296,514
R^2	0.539	0.542	0.545
Household FE	Yes	Yes	Yes
Month FE	Yes	Yes	
State-month trend		Yes	
State-month FE			Yes
Mean of outcome variable		0.048	

Notes: This table presents difference-in-differences estimates of the impact of DBTL policy on households' cooking-fuel expenditure share. The outcome variable household cooking fuel expenditure is constructed using CMIE household survey data (January to October 2014) covering the period during DBTL (*pre termination* = 1) and after its termination (*pre termination* = 0). On average, households spend 4.8 percent of their monthly expenditure on cooking fuel. Standard errors, clustered at the district level, are in parentheses.

to an increase in their cooking fuel expenditure share. Conversely, terminating the policy may lead to a decrease in their cooking fuel expenditure share. The available data permits only an estimation of the impact of terminating the DBTL policy.

Allowing for heterogeneous impact by total expenditure, Figure 9 shows the difference in cooking fuel expenditure share from the period when DBTL was terminated (March to October 2014) to the period when DBTL was mandatory (January–February 2014) using CMIE data. The DBTL policy had a statistically significant impact on cooking fuel expenditure share only in the top expenditure quintile. Households in the top quintile are less likely to be excluded from subsidized gas due to lack of bank account or Aadhaar, but they are more likely to opt voluntarily not to enroll in DBTL. The pattern shown in Figure 9 provides at least partial evidence in support of richer households' self-selection *out* of gas subsidies.

Impact across Initial Incidence of Benefits.—I assess relative progressivity by estimating the DBTL policy's impact on the number of subsidized household refills purchased, by initial gas purchase levels (i.e., gas refills purchased before DBTL) using data on all households who purchased at least one gas refill in 2013–2014. The median household purchased seven 14.2 kg gas refills per year, which would translate to less than three refills in the five-month pre-DBTL period. For the five-month period preceding the DBTL rollout in Phase 1 districts (April–August 2013), the pro-rated cap would be four subsidized gas refills (based on an annual cap of nine per household account).

I find a progressive effect, in that the impact of DBTL rollout is significantly greater for household accounts with a higher initial incidence of benefit before DBTL than for those with low initial incidence (Figure 10). Columns 1 and 2 in online Appendix Table D.9 show that household accounts purchasing more than four gas refills in the pre-DBTL period observed a 69 percent and 84 percent higher effect of DBTL than household accounts purchasing up to four gas refills, respectively. As shown in

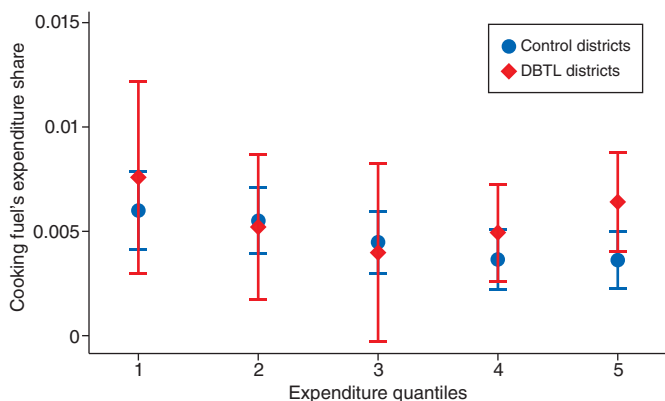


FIGURE 9. DBTL'S IMPACT ON COOKING FUEL EXPENDITURE

Notes: This plot shows the impact of DBTL policy on household cooking fuel's expenditure share across monthly total expenditure quintiles. It employed a triple-interaction (i.e., $DBTL \times Pre\ Termination \times Expenditure\ quintiles$) regression model. The outcome variable is cooking fuel's expenditure share. Household-month level data from CMIE surveys are used. The data covers January–October 2014, which includes two months (January and February 2014) of the DBTL enforcement period. Thus, the coefficient plot shows the change in fuel expense proportion between the DBTL enforcement period and after its termination (March–October 2014) for households in DBTL and non-DBTL districts. Ninety-five percent confidence interval with standard errors clustered at the district level is shown.

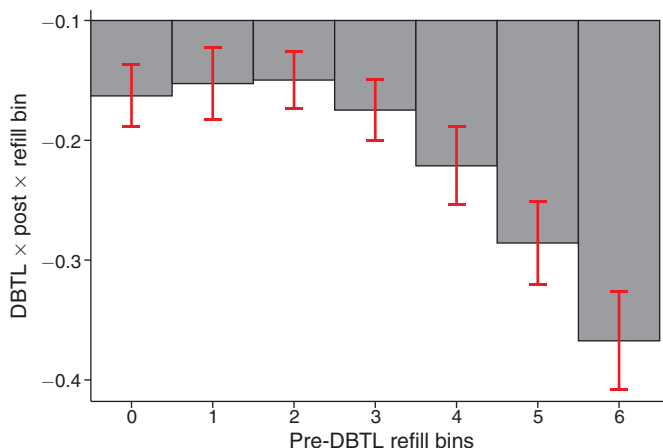


FIGURE 10. DISTRIBUTIONAL EFFECT ACROSS THE INITIAL SUBSIDY INCIDENCE

Notes: This graph shows the impact of DBTL policy on subsidized gas purchases across different bins of pre-DBTL subsidy incidence. Household accounts are categorized into seven groups starting with zero subsidized refills purchases to six or more refills purchased during the five months leading up to the DBTL rollout (April–August 2013). The median household account purchased three refills in this period. Household and month fixed effects are included. Ninety-five percent confidence intervals with standard errors clustered at the district level are shown.

Figure 4, more than 80 percent of accounts with low-to-normal initial usage (1–4 gas refills) enrolled in DBTL, while less than 50 percent of accounts with high initial usage (more than four gas refills) enrolled. Legitimate households that do have high-usage accounts in the pre-DBTL period are more likely to be high-income.

Figure 10 and online Appendix Table D.9 provide suggestive evidence that DBTL's impact was progressive in that it disproportionately reduced benefits to household accounts that initially benefited highly from the subsidy. Part of this may be explained by the fact that a significant proportion of high-usage accounts are ghost accounts, and hence cannot enroll in DBTL to avail of subsidized gas refills.

One important caveat is that a household gas account that purchased more refills in the pre-period had more room for a negative impact, and so at least part of the estimated differential effect could be mechanical. Online Appendix Table D.9 (columns 3 and 4) shows that there is no gradual increase in DBTL's impact on accounts with fewer pre-DBTL refills (three or less), while there is a gradual and significant increase in the impact on household accounts purchasing four or more gas refills in the five-month period preceding DBTL rollout. This mitigates the concern that these results are only due to mechanical factors.

F. Diversion of Fuel by Households

Does directly transferring subsidies to households cause an increase in gas subsidy diversion by genuine households?¹⁷ Many households are infra-marginal with respect to the annual cap for subsidized gas. The median household gas user purchases seven refills, which is well below the annual cap. An increase in the diversion by households in response to DBTL would counteract any negative effect on the diversion of household gas by ghost accounts, both in estimation and reality.¹⁸ Similarly, it may also mitigate the supply shock in the black market leading to a smaller estimated effect on black market prices. The estimated effects on household purchases and black market prices already net out any shift in diversion to genuine households. Still, from a fiscal savings perspective, it is important to understand whether there is any additional reselling by households in response to the higher black market prices.

I test for diversion by households by comparing gas purchases made by legitimate household accounts in DBTL districts and in districts where DBTL was yet to be enforced. Households who enrolled in DBTL at any point are presumably legitimate households, given the Aadhaar requirement. Table 10 (column 1) shows that enforcing DBTL caused a 4 percent increase in purchases of gas refills by households in treated districts when compared with households in voluntary DBTL districts (Phases 3–6). Despite the observed increase, the magnitude of the coefficient is relatively small compared to the overall impact on household gas purchases. Column 2 shows that there is no change in gas purchases due to DBTL when only households in Phase 6 are included in the control group. Since Phase 6 districts did not enter the voluntary DBTL period until January 2014, restricting the control group to Phase 6 allays concerns about voluntary DBTL enrollment affecting household gas purchases. Online Appendix Figure D.7 shows that, in general, gas purchases by

¹⁷ Related research on tax evasion suggests policies to reduce evasion may lead to a displacement in evasion, as long as economic incentives remain in place (Yang 2008; Mishra, Subramanian, and Topalova 2008; Jose, Cadot, and Olarreaga 2006).

¹⁸ Reselling of household gas by legitimate households in the black market is qualitatively different from the leakage through ghost accounts. Especially when primarily low-income households are engaged in gas reselling, this may even be a way to enhance redistribution. However, as per the law, unauthorized persons cannot sell or resell gas refills.

TABLE 10—IMPACT OF DBTL ON GAS PURCHASES BY HOUSEHOLDS ENROLLED IN DBTL

	Number of LPG refills purchased	
	(1)	(2)
<i>DBTL × post</i>	0.0234 (0.00492)	−0.00195 (0.00562)
Observations	9,542,478	3,899,918
Household FE	Yes	Yes
Month FE	Yes	Yes
Control districts	Phase 3–6	Phase 6
Mean of outcome variable	0.546	0.551

Notes: This table reports difference-in-differences estimates for the impact of the DBTL policy on gas purchases by household accounts enrolled in DBTL. Household-by-month level panel data covering the time period before and during the DBTL policy reform (April 2013 to February 2014) is used. The outcome variable is the number of gas refills purchased by a household account in a given month. All household accounts enrolled in DBTL by March 2014 are included. Household accounts that did not enroll in DBTL by March 2014 are not included in this sample. Column 1 includes households voluntarily enrolled in DBTL from Phase 3–6 districts as the control group. Column 2 includes households voluntarily enrolled in DBTL from Phase 6 districts as the control group. Standard errors are in parentheses. Standard errors are clustered at the district level.

legitimate household accounts under DBTL were not significantly higher compared to the counterfactual in five out of six months during DBTL. These results suggest that there was no meaningful increase in diversion by legitimate household accounts once the DBTL policy was implemented, and, overall, diversion by households does not seem to outweigh the reduction in diversion by suppliers and other agents.

There are two potential explanations for why, despite the high prices households could fetch on the black market, they do not increase their engagement in the diversion. First, households may face higher legal and logistical costs when selling gas refills on the black market, especially since they do not have access to aggregation technology or the market access that suppliers do. Second, social norms may make it costly for a household to participate in the black market.

G. DBTL Reimplementation across the Country

The newly elected Indian government reimplemented DBTL in 54 districts on November 15, 2014, and in the remaining 622 districts from January 1, 2015. Table 11 shows difference-in-differences estimates including data on purchases from one full month (i.e., December 2014) following the re-rollout to 54 districts in the first phase but before the rollout to the remaining districts. Household gas purchases show a pattern similar to that reported in Table 4, with a 13 to 18 percent decrease (columns 1–3). Commercial gas purchases increased by 11 to 13 percent (columns 4–6). In the months following DBTL implementation, commercial 19kg gas refill sales across the country increased by more than 50 percent, compared with the flat sales in the pre-period (online Appendix Figure D.5).¹⁹

¹⁹ A similar trend is observed in the amount of gas sold as a transportation fuel, following DBTL's country-wide implementation (Jain, Agrawal, and Ganeshan 2018).

TABLE 11—IMPACT OF DBTL REIMPLEMENTATION ON HOUSEHOLD AND COMMERCIAL GAS PURCHASES

	Panel A. $\log(\text{household LPG refills})$			Panel B. $\log(\text{commercial LPG refills})$		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>DBTL2</i> $\times$ <i>post</i>	−0.180 (0.0158)	−0.171 (0.0161)	−0.128 (0.0207)	0.111 (0.0314)	0.117 (0.0334)	0.127 (0.0385)
Observations	99,369	99,369	99,362	77,193	77,193	77,193
R^2	0.956	0.956	0.957	0.930	0.930	0.931
Distributor FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes		Yes	Yes	
State-month trend		Yes			Yes	
State-month FE			Yes			Yes

Notes: This table reports the difference-in-differences estimates of the impact of DBTL reimplementation in November 2014. The DBTL2 group includes the 54 Phase 1 and Phase 2 districts, whereas the control group includes all remaining districts. This regression uses data from May to December 2014. Data for November is omitted because the DBTL policy was implemented in the middle of November 2014. Columns 1–3 report the estimated difference-in-differences coefficient for household LPG, while columns 4–6 report the same for commercial LPG. Standard errors, clustered at the district level, are in parentheses.

V. Conclusion

More than 80 percent of countries across the world provide in-kind transfers, with low-income countries in particular spending a high proportion of their total safety net budget on in-kind transfer programs (World Bank 2015). How developing countries can improve in-kind transfer programs is an important question with significant welfare implications. This paper studies a policy reform in a large, in-kind transfer program in India that sought to bridge the over-the-counter price gap by transferring the subsidy amount directly to beneficiaries. The results show that this policy change led to a significant reduction in subsidized gas sales to household accounts. This is followed by increased gas sales to commercial accounts on the legal market, which indicates that a significant part of this reduction in household sales represented a decrease in the leakage of subsidized household gas to the black market. Furthermore, the sudden termination of the policy offers consistent empirical evidence regarding its impact. Specifically, survey data reveals a decline in the black market price of gas following the termination of the policy. It provides additional substantiation for the claim that the DBTL policy curbed the diversion of subsidized gas. There is little evidence that diversion displaced to new channels (e.g., legitimate households purchasing more gas to resell in the black market). Empirical analysis shows that households did not increase their gas purchases significantly, at least in the short run, and the increase in commercial gas purchases was sustained over time. Overall, these findings support the narrative that the arbitrage incentives created by a dual-pricing design of in-kind transfers led to a high level of diversion of public funds in the gas subsidy program.

Taken as a whole, the empirical evidence provided in this study supports policy reforms aimed at moving away from the traditional dual-price implementation of in-kind transfer programs, such as public distribution of subsidized fuel, food, and fertilizer, to direct transfers. India spent roughly US\$ 14 billion on household gas

subsidies from 2015 to 2019, potentially saving policymakers between US\$3 billion and US\$6 billion compared to estimated expenditures in the absence of DBTL policy. This is due to the estimated reduction of about 17–19 percent in total household gas and 46 percent in subsidized household gas. Increased tax revenues from commercial gas sales further strengthen the case. The cost of direct bank transfers appears to be minor (fixed at a 1 percent fee to banks by the central government) relative to potential gains.

The gains from DBTL must be weighed against any welfare loss due to the exclusion of legitimate households. Lacking matched primary household survey data and administrative records, this study does not attempt to provide estimates of exclusion. Results suggest that DBTL did not increase the household cooking fuel expenditure share on average or across the expenditure quintile, except in the top quartile. It is worth reiterating that, during the period of this study, gas was used primarily by middle- and higher-income households in India. Since the implementation of DBTL, India has expanded access to cooking gas to almost 100 million low-income households with an additional startup subsidy. Importantly, short-term liquidity constraints may affect low-income households' decision to use subsidized gas when they are required to pay the full price of gas upfront. More research is needed to understand the importance of such factors for low-income households in the context of policy reforms using income transfers.

On a broader level, reforms to welfare delivery need much stronger political support, especially when adversely affected groups are likely to lobby against (or subvert) such reforms. The abrupt termination of DBTL before the 2014 general elections likely reflected exactly these considerations. Casaburi and Troiano (2016) show that incumbent politicians may enjoy a higher likelihood of reelection when more vigorously enforcing an anti-tax evasion policy. However, such potential gains may not be visible to the politicians themselves *ex ante*, especially in developing countries.

REFERENCES

- Aji, Sowmya. 2012. "Stink of a Gas Scam: Over 24 Lakh of the 79 Lakh LPG Connections in Karnataka Belong to 'Ghost' Consumers." *India Today*, March 26. <https://www.indiatoday.in/magazine/nation/story/20120326-karnataka-24-lakh-fake-domestic-gas-connections-uncovered-757753-2012-03-17>.
- Alderman, Harold. 2002. "Subsidies as a Social Safety Net: Effectiveness and Challenges." Unpublished.
- Alesina, Alberto. 1997. *Political Cycles and the Macroeconomy*. Cambridge, MA: MIT Press.
- Arze del Granado, Francisco Javier, David Coady, and Robert Gillingham. 2012. "The Unequal Benefits of Fuel Subsidies: A Review of Evidence for Developing Countries." *World Development* 40 (11): 2234–48.
- Balakrishnan, Pulapre, and Bharat Ramaswami. 1997. "Quality of public Distribution System: Why It Matters." *Economic and Political Weekly* 32 (4): 162–65.
- Banerjee, Abhijit, Esther Duflo, Clement Imbert, Santhosh Mathew, and Rohini Pande. 2020. "E-governance, Accountability, and Leakage in Public Programs: Experimental Evidence from a Financial Management Reform in India." *American Economic Journal: Applied Economics* 12 (4): 39–72.
- Banerjee, Abhijit, Rema Hanna, Jordan Kyle, Benjamin A. Olken, and Sudarno Sumarto. 2018. "Tangible Information and Citizen Empowerment: Identification Cards and Food Subsidy Programs in Indonesia." *Journal of Political Economy* 126 (2): 451–91.
- Banerjee, Abhijit, Rema Hanna, Benjamin A. Olken, Elan Satriawan, and Sudarno Sumarto. 2023. "Electronic Food Vouchers: Evidence from an At-Scale Experiment in Indonesia." *American Economic Review* 113 (2): 514–47.

- Banerjee, Abhijit, and Rohini Pande.** 2007. "Parochial Politics: Ethnic Preferences and Politician Corruption." Unpublished.
- Barnwal, Prabhat.** 2024. *Data and code for: "Curbing Leakage in Public Programs: Evidence from India's Direct Benefit Transfer Policy."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E195367V1>.
- Beegle, Kathleen, Aline Coudouel, and Emma Monsalve.** 2018. *Realizing the Full Potential of Social Safety Nets in Africa*. Washington, DC: World Bank Publications.
- Besley, Timothy, and Stephen Coate.** 1991. "Public Provision of Private Goods and the Redistribution of Income." *American Economic Review* 81 (4): 979–84.
- Burgess, Robin, Matthew Hansen, Benjamin A. Olken, Peter Potapov, and Stefanie Sieber.** 2012. "The Political Economy of Deforestation in the Tropics." *Quarterly Journal of Economics* 127 (4): 1707–54.
- Bussell, Jennifer L.** 2010. "Why Get Technical? Corruption and the Politics of Public Service Reform in the Indian States." *Comparative Political Studies* 43 (10): 1230–57.
- Casaburi, Lorenzo, and Ugo Troiano.** 2016. "Ghost-House Busters: The Electoral Response to a Large Anti-Tax Evasion Program." *Quarterly Journal of Economics* 131 (1): 273–314.
- Chakravarty, Sayantan, and Charulata Joshi.** 1997. "Ex-Petroleum Minister Satish Sharma and his Beneficiaries Come under CBI Scrutiny." *India Today*, April 15. <https://www.indiatoday.in/magazine/nation/story/19970415-ex-petroleum-minister-satish-sharma-and-his-beneficiaries-come-under-cbi-scrutiny-831260-1997-04-14>.
- Cole, Shawn.** 2009. "Fixing Market Failures or Fixing Elections? Agricultural Credit in India." *American Economic Journal: Applied Economics* 1 (1): 219–50.
- Deshpande, Manasi, and Yue Li.** 2019. "Who is Screened Out? Application Costs and the Targeting of Disability Programs." *American Economic Journal: Economic Policy* 11 (4): 213–48.
- Economic Times.** 2016. "Give-it-Up: Over 1 crore LPG users gave up their subsidies." *Economic Times*, April 21. <https://economictimes.indiatimes.com/industry/energy/oil-gas/give-it-up-over-1-crore-lpg-users-gave-up-their-subsidies/articleshow/51929960.cms?from=mdr>.
- Finkelstein, Amy, and Matthew J. Notowidigdo.** 2019. "Take-up and Targeting: Experimental Evidence from SNAP." *Quarterly Journal of Economics* 134 (3): 1505–56.
- Fisman, Raymond, and Shang-Jin Wei.** 2004. "Tax Rates and Tax Evasion: Evidence from "Missing Imports" in China." *Journal of Political Economy* 112 (2): 471–96.
- Fisman, Raymond, and Shang-Jin Wei.** 2009. "The Smuggling of Art, and the Art of Smuggling: Uncovering the Illicit Trade in Cultural Property and Antiques." *American Economic Journal: Applied Economics* 1 (3): 82–96.
- Foremny, Dirk, and Nadine Riedel.** 2014. "Business Taxes and the Electoral Cycle." *Journal of Public Economics* 115: 48–61.
- Gadenne, Lucie.** 2018. "Can Rationing Increase Welfare? Theory and an Application to India's Ration Shop System." Unpublished.
- Gould, Carlos F., and Johannes Urpelainen.** 2018. "LPG as a Clean Cooking Fuel: Adoption, Use, and Impact in Rural India." *Energy Policy* 122: 395–408.
- Hota, A. P.** 2014. "Why Suspending DBT Scheme Would be a Retrograde Step." *Economics Times*, February 19. <https://economictimes.indiatimes.com/why-suspending-dbt-scheme-would-be-a-retrograde-step/articleshow/30644333.cms?from=mdr>.
- IIPS and ICF.** 2017. *National Family Health Survey (NFHS-4), 2015-2016: India*. Mumbai: International Institute for Population Sciences.
- IISD.** 2014. *Indonesia Energy Subsidy Review*. Winnipeg, Canada: IISD.
- Jacoby, Hanan G.** 1997. "Self-Selection and the Redistributive Impact of In-Kind Transfers: An Econometric Analysis." *Journal of Human Resources* 32 (2): 233–49.
- Jain, Abhishek, Shaly Agrawal, and Karthik Ganesan.** 2018. "Lessons from the World's Largest Subsidy Benefit Transfer Scheme." In *Politics of Fossil Fuel Subsidies and Their Reform*, edited by Jakob Skovgaard and Harro van Asselt, 212–28. Cambridge, UK: Cambridge University Press.
- Jai, Shreya.** 2013. "Distributors Want Licence to Sell Cylinders in Black." *Economic Times*, July 30. <https://economictimes.indiatimes.com/industry/energy/oil-gas/distributors-want-licence-to-sell-cylinders-in-black/articleshow/21473130.cms?from=mdr>.
- Jose, Anson, Olivier Cadot, and Marcelo Olarreaga.** 2006. "Tariff Evasion and Customs Corruption: Does Pre-Shipments Inspection Help?" *B.E. Journal of Economic Analysis and Policy* 5 (1): 1–24.
- Kapur, Devesh, and Milan Vaishnav.** 2011. "Quid Pro Quo: Builders, Politicians, and Election Finance in India." Unpublished.

- Kar, Abhishek, Shonali Pachauri, Rob Bailis, and Hisham Zerriffi. 2019. "Using Sales Data to Assess Cooking Gas Adoption and the Impact of India's Ujjwala Programme in Rural Karnataka." *Nature Energy* 4 (9): 806–14.
- Khera, Reetika. 2011a. "Trends in Diversion of Grain from the Public Distribution System." *Economic and Political Weekly* 46 (21): 106–14.
- Khera, Reetika. 2011b. "The UID Project and Welfare Schemes." *Economic and Political Weekly* 46 (9): 38–43.
- Kleven, Henrik Jacobsen, and Wojciech Kopczuk. 2011. "Transfer Program Complexity and the Take-Up of Social Benefits." *American Economic Journal: Economic Policy* 3 (1): 54–90.
- Kojima, Masami. 2013. "Petroleum Product Pricing and Complementary Policies: Experience of 65 Developing Countries since 2009." Unpublished.
- Kopczuk, Wojciech, Justin Marion, Erich Muehlegger, and Joel Slemrod. 2016. "Does Tax-Collection Invariance Hold? Evasion and the Pass-Through of State Diesel Taxes." *American Economic Journal: Economic Policy* 8 (2): 251–86.
- Kotwal, Ashok, Milind Murugkar, and Bharat Ramaswami. 2011. "PDS Forever?" *Economic and Political Weekly* 46 (21): 72–76.
- Kumar, Manoj. 2012. "Cap on LPG Fuels Nationwide Rage, Distributors Call for Strike on Oct 1." *International Business Times*, September 24. <https://www.ibtimes.co.in/cap-on-lpg-fuels-nationwide-rage-distributors-call-for-strike-on-oct-1-387303>.
- Lahoti, Rahul, J.Y. Suchitra, and Prodyumna Goutam. 2012. "Subsidies for Whom: The Case of LPG in India." *Economic and Political Weekly* 47 (44): 16–18.
- Marion, Justin, and Erich Muehlegger. 2008. "Measuring Illegal Activity and the Effects of Regulatory Innovation: Tax Evasion and the Dyeing of Untaxed Diesel." *Journal of Political Economy* 116 (4): 633–66.
- McLure Jr, Charles E. 2014. "Reforming Subsidies for Fossil Fuel Consumption: Killing Several Birds with One Stone." *Taxation and Development: The Weakest Link?*, edited by Richard M. Bird and Jorge Martinez-Vazquez, 238–87. Cheltenham, UK: Edward Elgar Publishing Limited.
- Ministry of Petroleum and Natural Gas. 2011. *Indian Petroleum and Natural Gas Statistics 2010–2011*. New Delhi: Ministry of Petroleum and Natural Gas.
- Ministry of Petroleum and Natural Gas. 2014a. *Marketing Discipline Guidelines*. New Delhi: Ministry of Petroleum and Natural Gas.
- Ministry of Petroleum and Natural Gas. 2014b. *Report on Marketing Activities*. New Delhi: Ministry of Petroleum and Natural Gas.
- Ministry of Petroleum and Natural Gas. 2015. *Indian Petroleum and Natural Gas Statistics 2014–2015*. New Delhi: Ministry of Petroleum and Natural Gas.
- Ministry of Petroleum and Natural Gas. 2018. *Press Release on Transfer of Subsidy to Domestic LPG Consumers*. New Delhi: Ministry of Petroleum and Natural Gas.
- Mishra, Prachi, Arvind Subramanian, and Petia Topalova. 2008. "Tariffs, Enforcement, and Customs Evasion: Evidence from India." *Journal of Public Economics* 92 (10–11): 1907–25.
- Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar. 2016. "Building State Capacity: Evidence from Biometric Smartcards in India." *American Economic Review* 106 (10): 2895–2929.
- Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar. 2023. "Identity Verification Standards in Welfare Programs: Experimental Evidence from India." *Review of Economics and Statistics*. https://doi.org/10.1162/rest_a_01296.
- Nagavarapu, Sriniketh, and Sheetal Sekhri. 2016. "Informal Monitoring and Enforcement Mechanisms in Public Service Delivery: Evidence from the Public Distribution System in India." *Journal of Development Economics* 121: 63–78.
- Nambiar, Nisha. 2014. "LPG subsidy: Direct Benefit Transfer System Stands Scrapped from Mar 10." *Times of India*, March 7. <https://indianexpress.com/article/cities/pune/lpg-subsidy-direct-benefit-transfer-system-stands-scrapped-from-mar-10/>.
- Nordhaus, William D. 1975. "The Political Business Cycle." *Review of Economic Studies* 42 (2): 169–90.
- NSSO. 2010. *India National Sample Survey 2009–2010 (66th round) - Type 1 - Consumer Expenditure*. National Sample Survey Organization.
- NSSO. 2012. *India National Sample Survey 2011–2012 (68th round) - Type 1 - Consumer Expenditure*. National Sample Survey Organization.
- NSSO. 2015a. *Energy Sources of Indian Households for Cooking and Lighting, 2011–2012, NSS 68th round*. Ministry of Statistics and Programme Implementation.
- NSSO. 2015b. *Public Distribution System and Other Sources of Household Consumption, 2011–2012*. National Sample Survey Organization.

- Olken, Benjamin A.** 2007. "Monitoring Corruption: Evidence from a Field Experiment in Indonesia." *Journal of Political Economy* 115 (2): 200–49.
- Olken, Benjamin A., and Rohini Pande.** 2012. "Corruption in Developing Countries." *Annual Review of Economics* 4 (1): 479–509.
- Pissarides, Christopher A., and Guglielmo Weber.** 1989. "An Expenditure-Based Estimate of Britain's Black Economy." *Journal of Public Economics* 39 (1): 17–32.
- Registrar General of India.** 2011. *Houselisting and Housing Census Data: 2011*.
- Sastry, Anil Kumar.** 2012. "Illegal LPG Connections: Centre Cold to Karnataka Model." *Hindu*, October 7. <https://www.thehindu.com/news/national/karnataka/Illegal-LPG-connections-Centre-cold-to-Karnataka-model/article12547834.ece>.
- Shleifer, Andrei, and Daniel Treisman.** 2001. *Without a Map: Political Tactics and Economic Reform in Russia*. Cambridge, MA: MIT Press.
- Slemrod, Joel.** 2008. "Does it Matter Who Writes the Check to the Government? The Economics of Tax Remittance." *National Tax Journal* 61 (2): 251–75.
- Sukhtankar, Sandip.** 2012. "Sweetening the Deal? Political Connections and Sugar Mills in India." *American Economic Journal: Applied Economics* 4 (3): 43–63.
- TNN.** 2013. "'Aadhaar-hit' Gas Delivery Boys Begin Strike." *Times of India*, September 19. <https://timesofindia.indiatimes.com/city/hyderabad/aadhaar-hit-gas-delivery-boys-begin-strike/articleshow/22730794.cms>.
- TNN.** 2014. "Aadhaar Link Needed to Control Subsidy Leakage: Moily." *Times of India*, January 19. <https://timesofindia.indiatimes.com/india/aadhaar-link-needed-to-control-subsidy-leakage-moily/articleshow/29027442.cms>.
- Troncoso, Karin, and Agnes Soares da Silva.** 2017. "LPG Fuel Subsidies in Latin America and the Use of Solid Fuels to Cook." *Energy Policy* 107: 188–96.
- Unique Identification Authority of India (UIDAI).** 2014. "Unique Identification Authority of India Dashboard." Unique Identification Authority of India. <https://uidai.gov.in/en/>.
- Vyas, Mahesh.** 2017. "Consumer Pyramids Survey, 2014 [India]." ICPSR. <https://doi.org/10.3886/ICPSR36782.v2>.
- World Bank.** 2015. *The State of Social Safety Nets 2015*. Washington, DC: World Bank.
- Yang, Dean.** 2008. "Can Enforcement Backfire? Crime Displacement in the Context of Customs Reform in the Philippines." *Review of Economics and Statistics* 90 (1): 1–14.